



Database Systems

Lecture 8:

The Entity Relationship Model (Part 1)

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Motivation



- ER diagrams are widely used
- ER model is easy to learn
 - Much simpler than UML
- An ER diagram is a good communication tool
 - Talking the same language

Learning Objectives



- After completing this lecture, you will be able to:
 - Create non-trivial ER diagrams
 - Assess the quality of an ER diagram
 - Perform and explain the mapping of ER diagrams to relations
 - Use a particular ER notation properly

Agenda

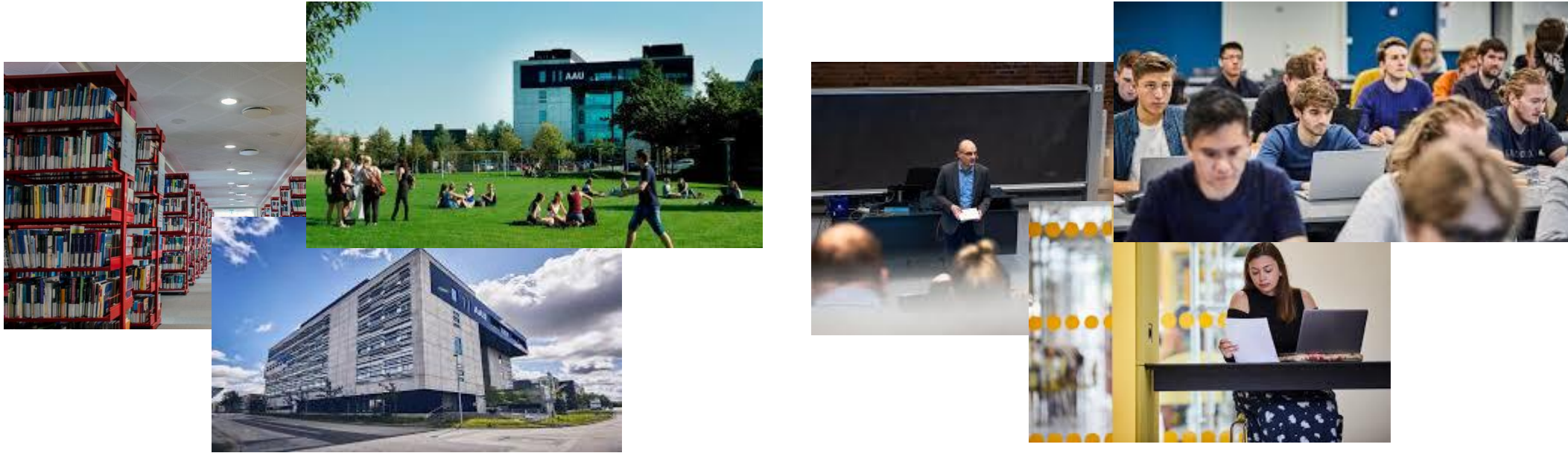


- Review Steps of Database Design
- ER Model
 - Basic Concepts
 - Mapping Cardinality Constraints
- Extended ER Model
 - Weak Entity Types
 - Specialization

Example Design – University Database



- Step 1: Requirement collection and analysis (What are we dealing with?)

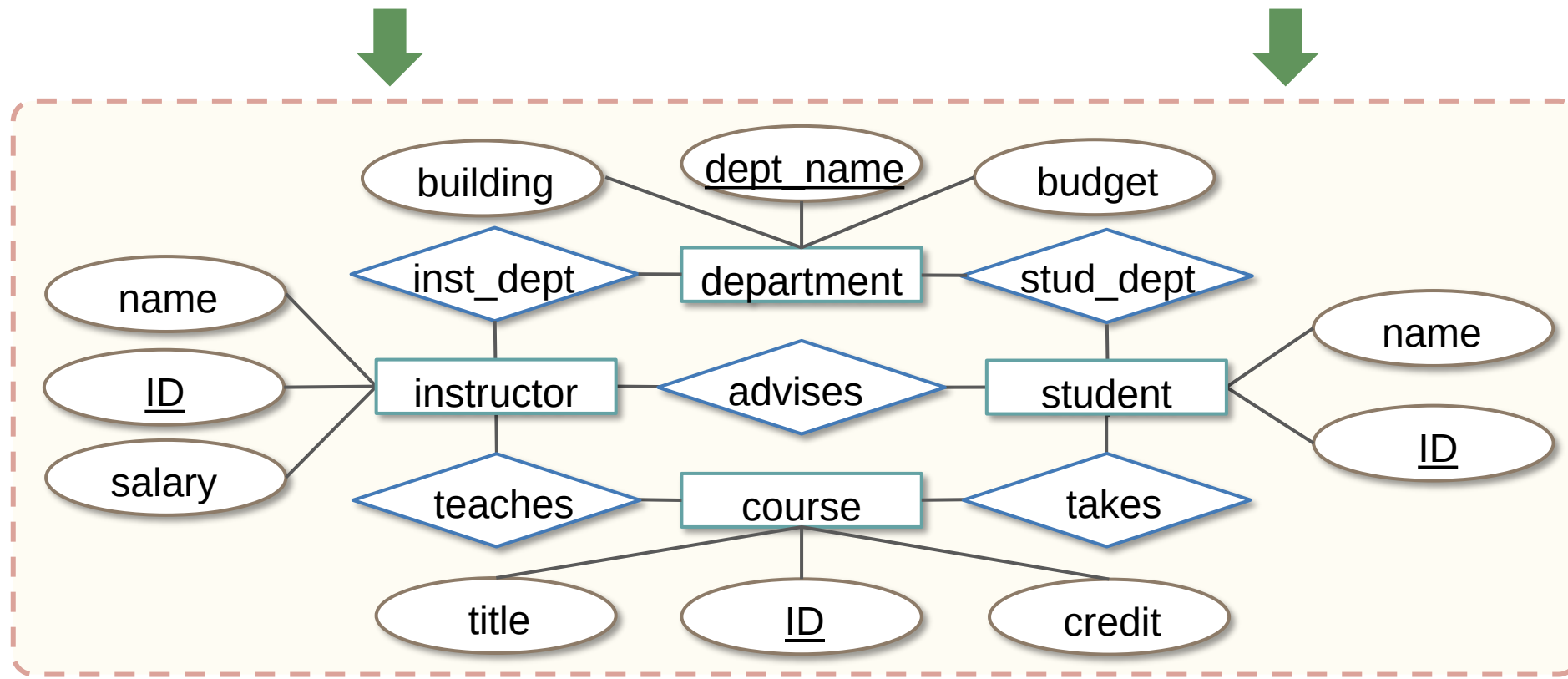


- “Instructors are in different departments”
- “Instructors teach courses”
- “Students take courses”
- ...

Example Design – University Database



- Step 2: Conceptual design (What data and relationships have to be captured?)
 - Requirements
 - ◆ “Instructors are in different departments”, “Instructors teach courses”, “Students take courses”, ...

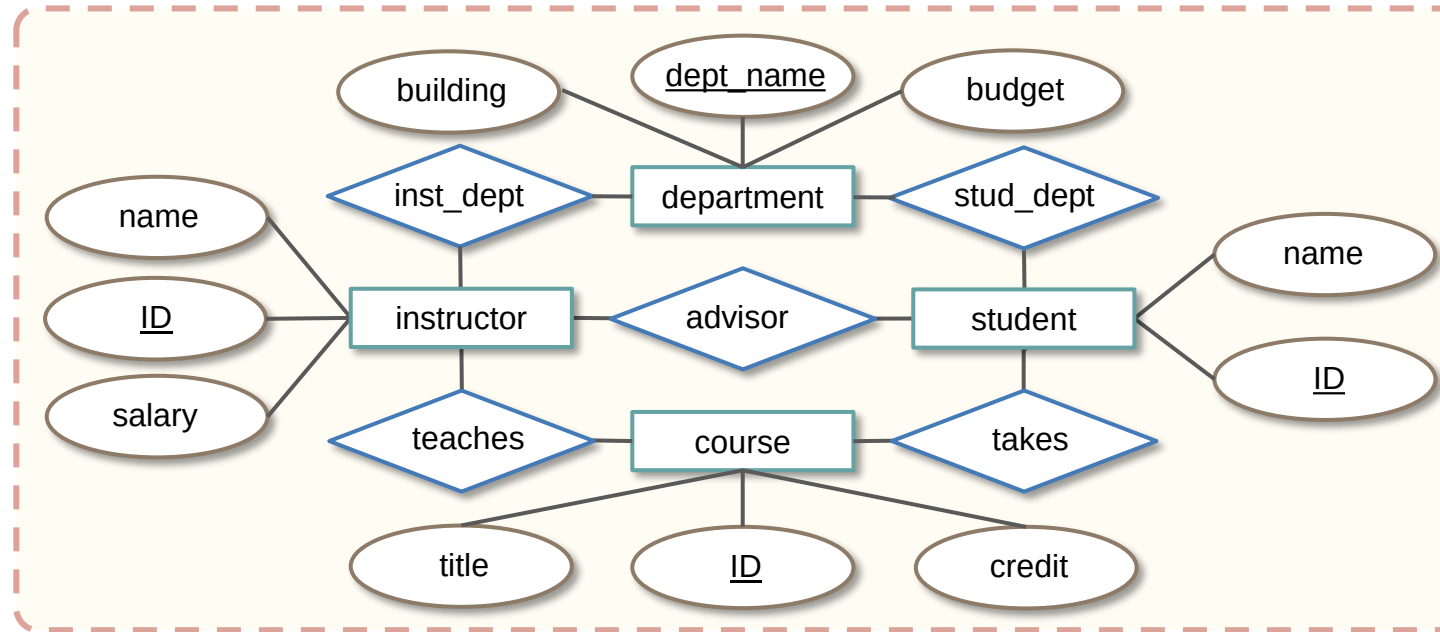


Easy for people to understand!

Example Design – University Database



- Step 3: Logical design (How to structure data in a specific model?)



- *instructor* (*ID*, *name*, *dept_name*, *salary*)
- *student* (*ID*, *name*, *dept_name*, *tot_cred*)
- *course* (*course_id*, *title*, *dept_name*, *credits*)
- *department* (*dept_name*, *building*, *budget*)

Easy for computer to understand!

Example Design – University Database



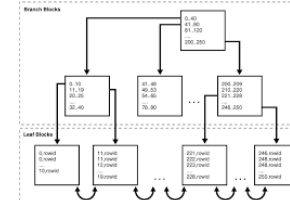
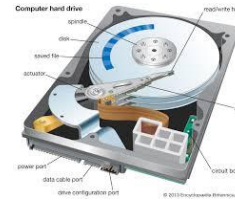
- Step 4: Physical Design (Which adaptations and optimizations does a specific DBMS require?)

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

<i>course_id</i>	<i>title</i>	<i>dept_name</i>	<i>credits</i>
BIO-101	Intro. to Biology	Biology	4
BIO-301	Genetics	Biology	4
BIO-399	Computational Biology	Biology	3
CS-101	Intro. to Computer Science	Comp. Sci.	4
CS-190	Game Design	Comp. Sci.	4
CS-315	Robotics	Comp. Sci.	3
CS-319	Image Processing	Comp. Sci.	3
CS-347	Database System Concepts	Comp. Sci.	3
EE-181	Intro. to Digital Systems	Elec. Eng.	3
FIN-201	Investment Banking	Finance	3
HIS-351	World History	History	3
MU-199	Music Video Production	Music	3
PHY-101	Physical Principles	Physics	4



Memory, pages, data structures, indexes, devices, ...

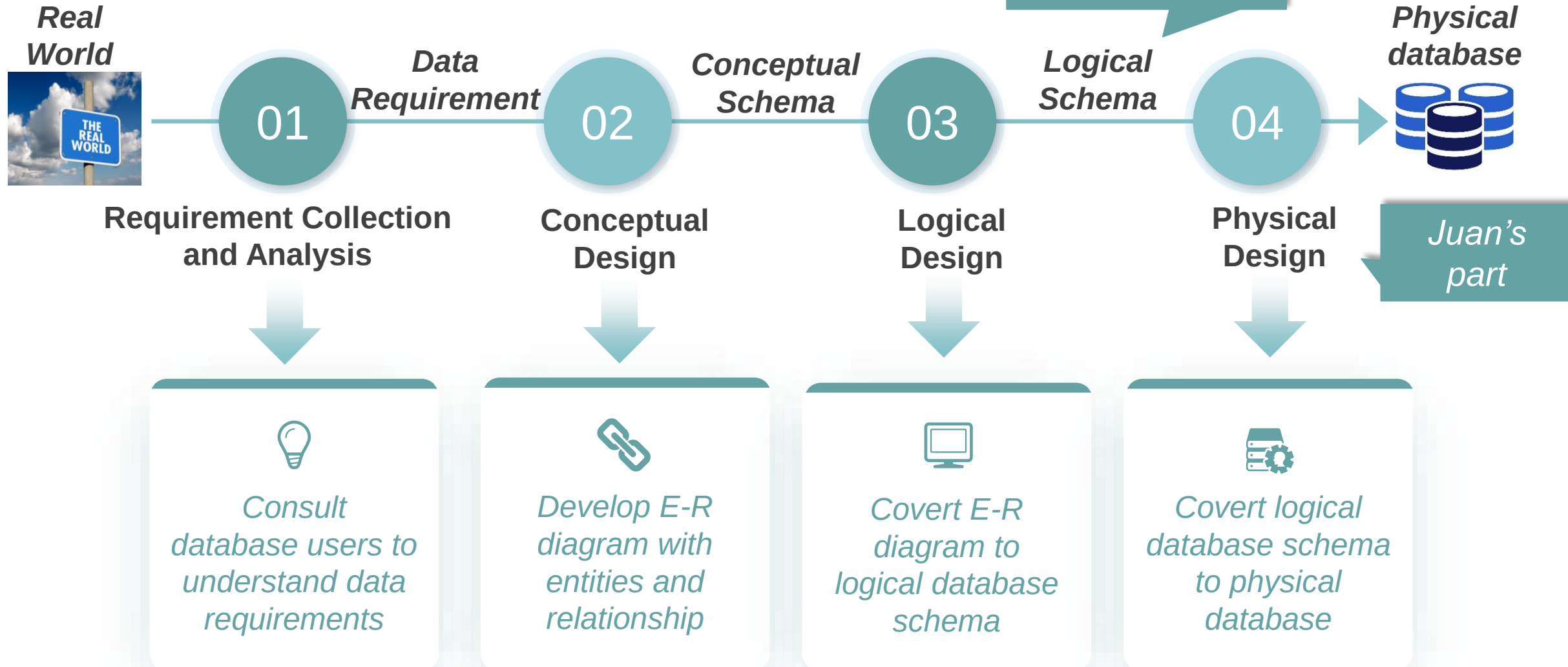


Steps of Database Design



Lecture 4

Relational table

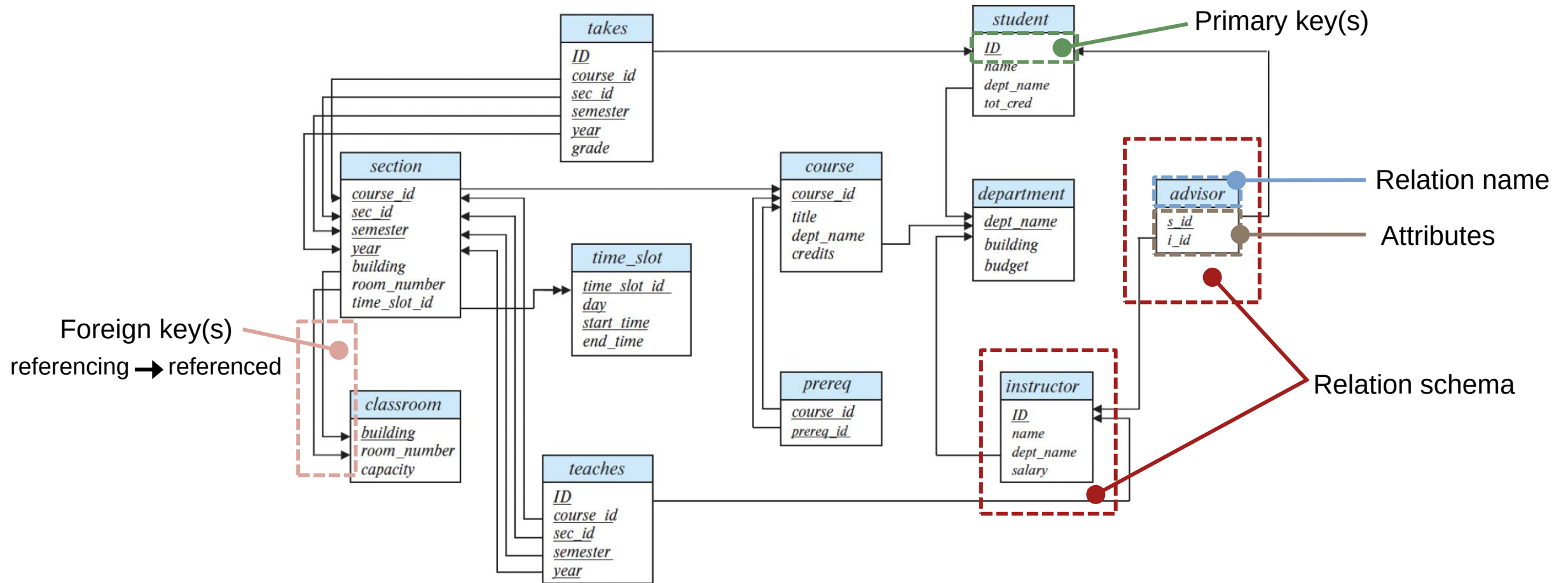


Physical Database Review



- Lecture 1-3: SQL
 - CREATE, ALTER, INSERT, DELETE, SELECT, JOIN
 - Logical vs. physical data view
- Lecture 5: Storage
 - Disk-based architecture, page and tuple layout
 - Row vs. column storage
 - Indexes: Hash, B+Tree, clustered / unclustered
- Lecture 6: Query Execution
 - From SQL to relational algebra and plan
 - Sorting, grouping, join algorithms
 - Cost mainly from I/O
- Lecture 7: Transactions
 - ACID: Atomicity, Consistency, Isolation, Durability
 - Serializability, locks, isolation levels

Lecture 4: Relational Model



Lecture 4: Relational Model



- Relational Algebra: A mathematical way to formulate queries on relations (i.e., tables).
 - Input: one or multiple relations
 - Output: a new relation

- Fundamental operations

- Selection σ
- Projection π
- Rename ρ



Unary operations
Operate on one relation

- Cartesian product $\times$
- Union $\cup$
- Difference $-$



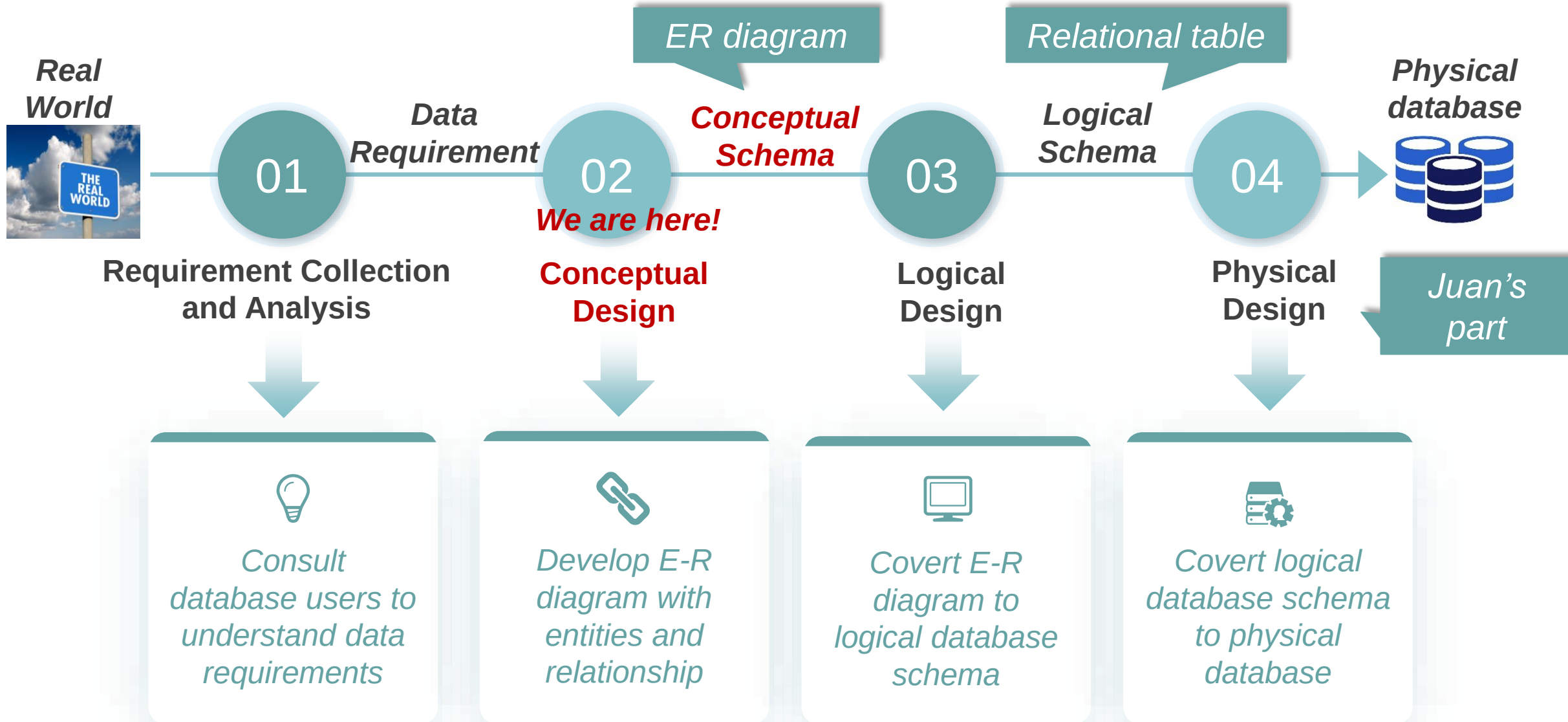
Binary operations
Operate on pairs of relations

- ✓ Any relational algebra query can be expressed with the set of fundamental operations only.
- ✓ Removing any one of these operations reduces the expressive power.
- ✓ Operations can be combined (with some rules)

Steps of Database Design



Lecture 4



Design Alternatives



- In designing a database schema, we must ensure that we avoid two major pitfalls:
 - Redundancy: a bad design may result in repeat information.
 - ◆ Redundant representation of information may lead to data inconsistency among the various copies of information
 - Incompleteness: a bad design may make certain aspects of the enterprise difficult or impossible to model.
- Avoiding bad designs is not enough.
 - ✓ There may be a large number of good designs from which we must choose.

Agenda

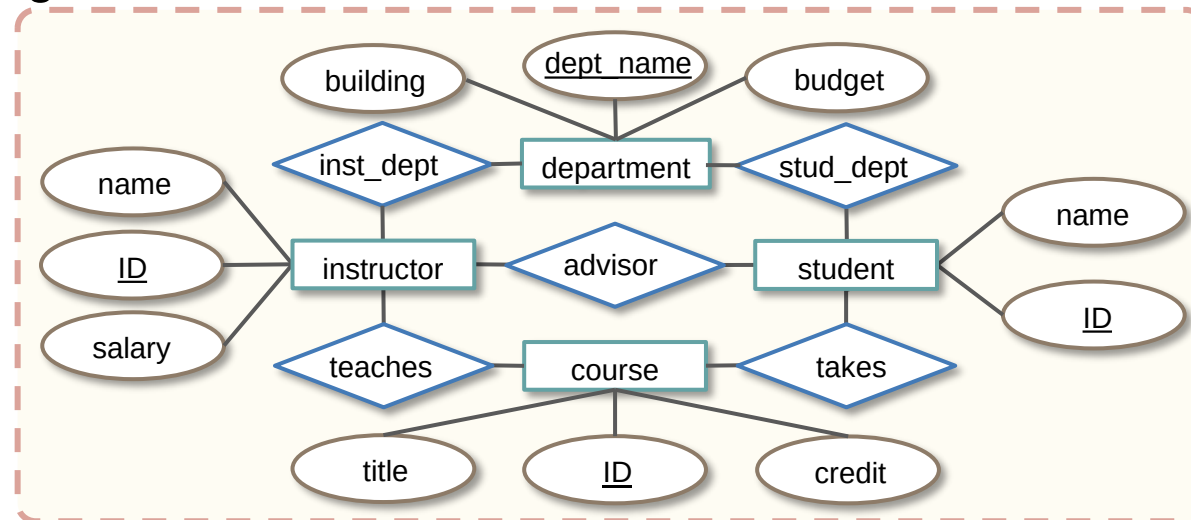


- Review Steps of Database Design
- Entity Relationship (ER) Model
 - Basic Concepts
 - Mapping Cardinality Constraints
- Extended ER Model
 - Weak Entity Types
 - Specialization

Entity Relationship (ER) Model



- The ER model is used in conceptual design
 - The second step of the data modelling process.
- The ER model employs three basic concepts:
 - Entity sets
 - Relationship sets
 - Attributes
- **ER diagram** provides a graphical representation of the ER model
 - It shows the overall logical structure of a database



An example of ER diagram

Entity Sets



- Concepts
 - An **entity** is an object that exists and is distinguishable from other objects.
 - ◆ Example: specific person, company, event, plant
 - An **entity set** is a set of entities of the same type that share the same properties.
 - ◆ Example: set of all persons, companies, trees, holidays
 - An **entity type** describes the entity set.
 - ◆ Example: *instructor* is an entity type with attributes: *ID*, *name*, *salary*

- ER diagram notation

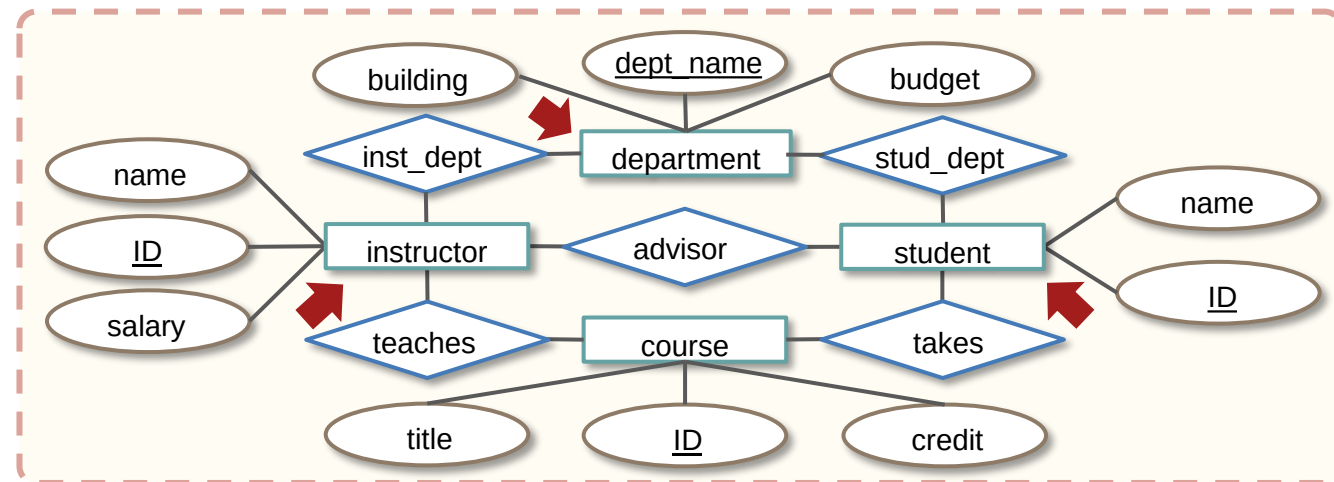
- Entity types → rectangles

- Example:

➡ *department*

➡ *instructor*

➡ *student*



Entity Sets



- How do we find entities when modeling?
 - Look for **nouns** in the requirements text
- Example — Video Store Database
 - Requirements
 - ◆ “The store rents video tapes of films to customers.”
 - ◆ “For each film, the store maintains a number of tapes.”
 - Nouns (entity candidates)
 - ◆ Customer
 - ◆ Film
 - ◆ Video tape

Customer

Film

Video tape

Relationship Sets

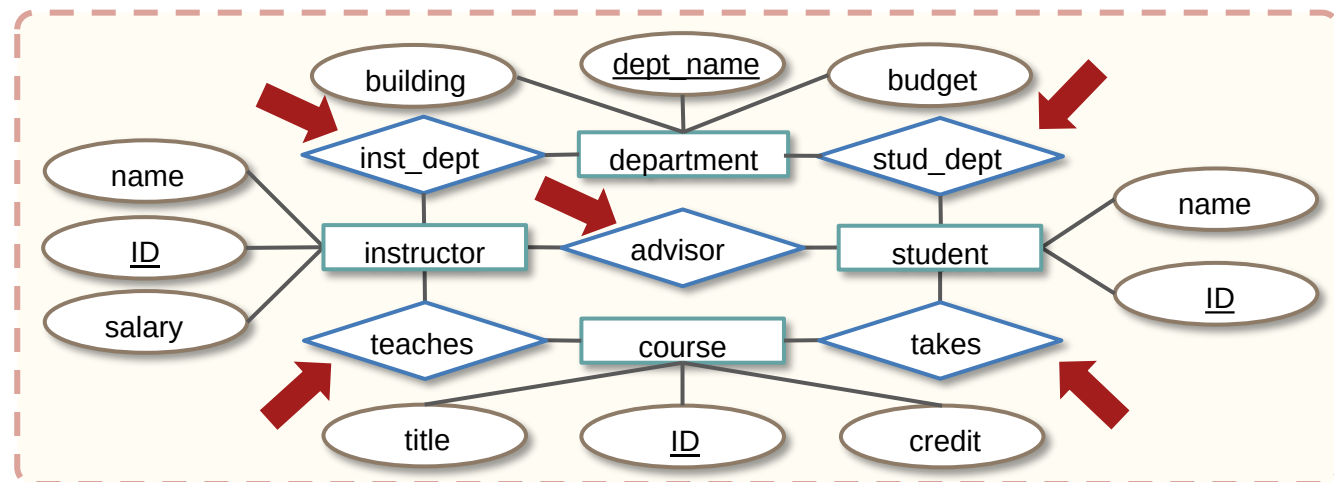


- Concepts
 - A **relationship** is an association among several entities
 - ◆ Example: a relationship *Publishes* associates the company Ubisoft with the game Assassin's Creed Shadows
 - A **relationship set** is a collection of relationship instances.
 - ◆ Example: {(Ubisoft, Assassin's Creed Shadows), (EA Sports, EA Sports FC 25), (Mojang, Minecraft), (Epic Games, Fortnite)} -- all the actual "publishing" links between specific companies and the games they release
 - A **relationship type** describes a relationship set.
 - ◆ Example: Publishes(company, game) -- the general association between video game companies and the games they publish

- ER diagram notation

- Relationship types → diamonds
- Example:

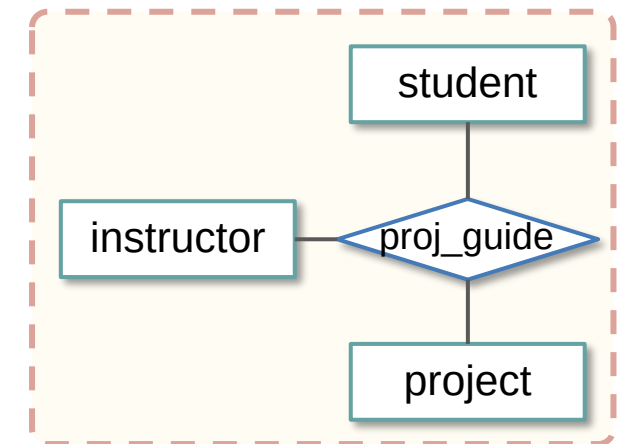
➡ *advisor*
➡ *inst_dept*
➡ *stud_dept*
➡ *teaches*
➡ *takes*



Relationship Sets



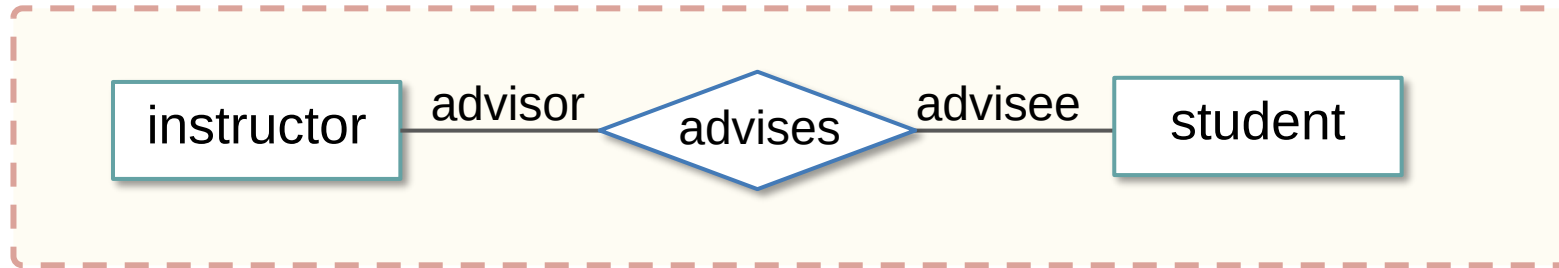
- Degree: Number of entity types that participate in a relationship.
- Binary relationship
 - Involve two entity sets (or degree two).
 - **Most** relationship sets in a database system are binary.
- Ternary relationship
 - Involve three entity sets (or degree three)
 - Relationships between more than two entity sets are **rare**.
- Example:
 - Students work on research projects under the guidance of an instructor.
 - Relationship *proj_guide* is a ternary relationship between instructor, student, and project
- In general: n-ary or n-way (multiway relationship types)



Roles in Relationships



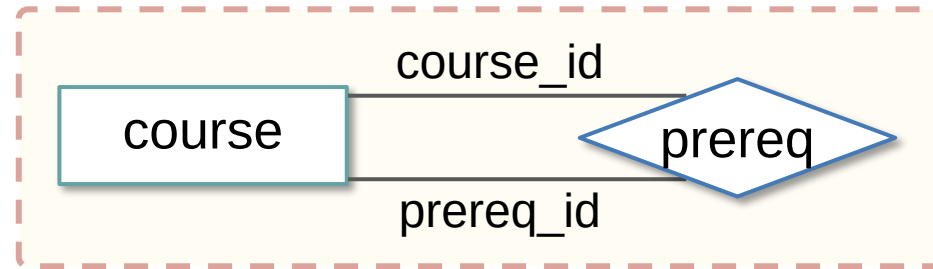
- Role names are optional and used to characterize a relationship type.
- Example:
 - Consider an instructor and student relationship:
 - ◆ Instructor: advisor
 - ◆ Student: advisee



Relationship Sets



- Recursive relationship
 - A relationship where the same entity set participates **more than once**
 - The entity **plays different roles** in the relationship
- Example:
 - Consider a course and its prerequisite relationship:
 - ◆ A course may require another course as its prerequisite.



- ◆ The two edges are labeled with `course_id` and `prereq_id`
- ◆ These labels are role names, distinguishing the two different roles that course plays in the relationship

Attributes



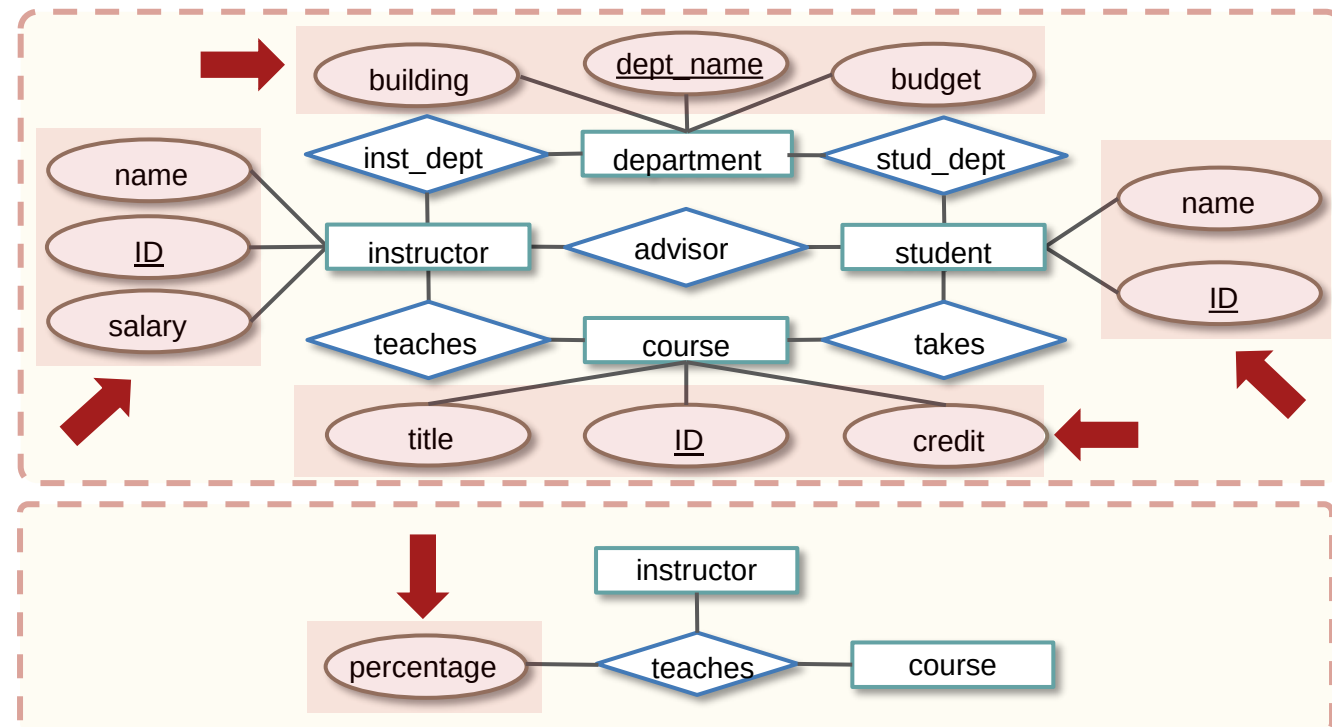
- Concepts
 - **Attributes** model characteristics of entities or relationships.
 - ◆ All entities of an entity type have the same characteristics
 - ◆ Attributes are declared for entity types
 - ◆ Attributes have a domain or value set
 - ◆ Relationship types can also have (descriptive) attributes.

- ER diagram notation

- Attributes → ovals

- Example:

→ *instructor.name*
→ *instructor.id*
→ *instructor.salary*
→ *department.dept_name*
→ *department.building*
→ *department.budget*
→ *teaches.percentage*
→ ...

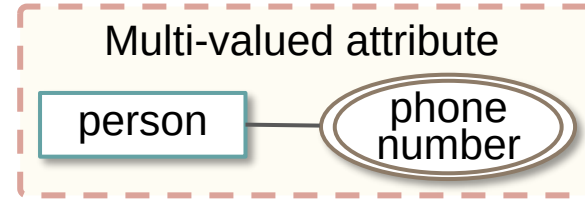
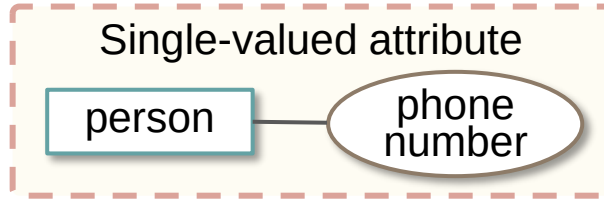


Attributes



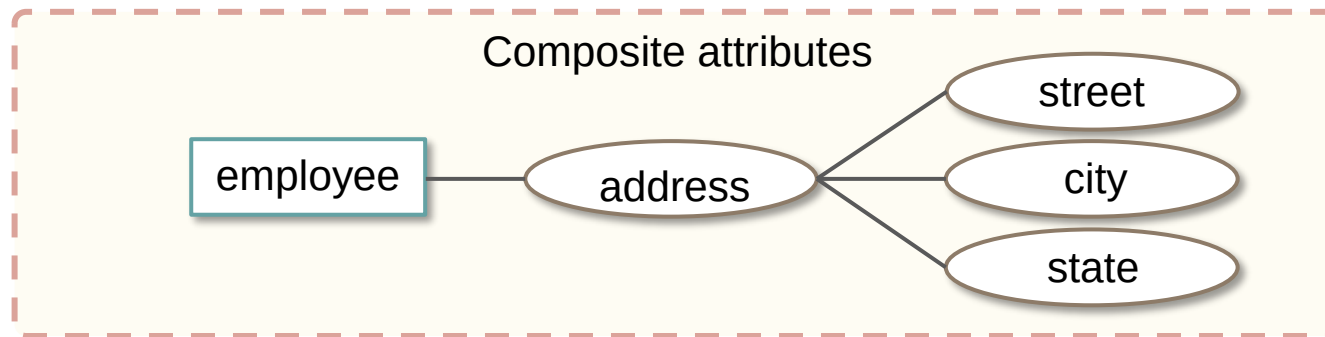
- Single-valued vs. multi-valued attributes

- A person might have multiple phone numbers (or a single one)



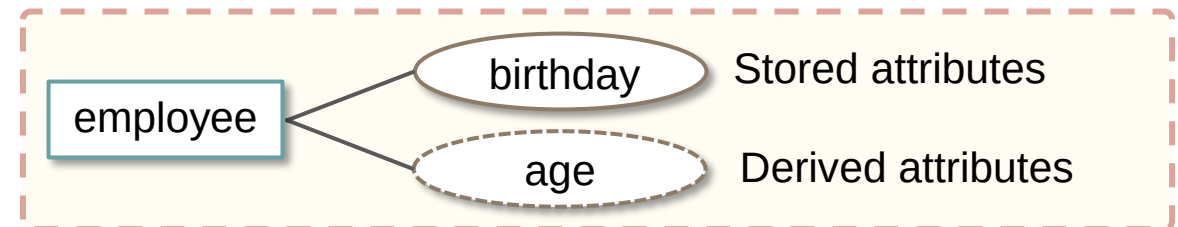
- Simple attributes vs. composite attributes

- An address can be modelled as a string or composed of street, city and state



- Stored attributes vs. derived attributes

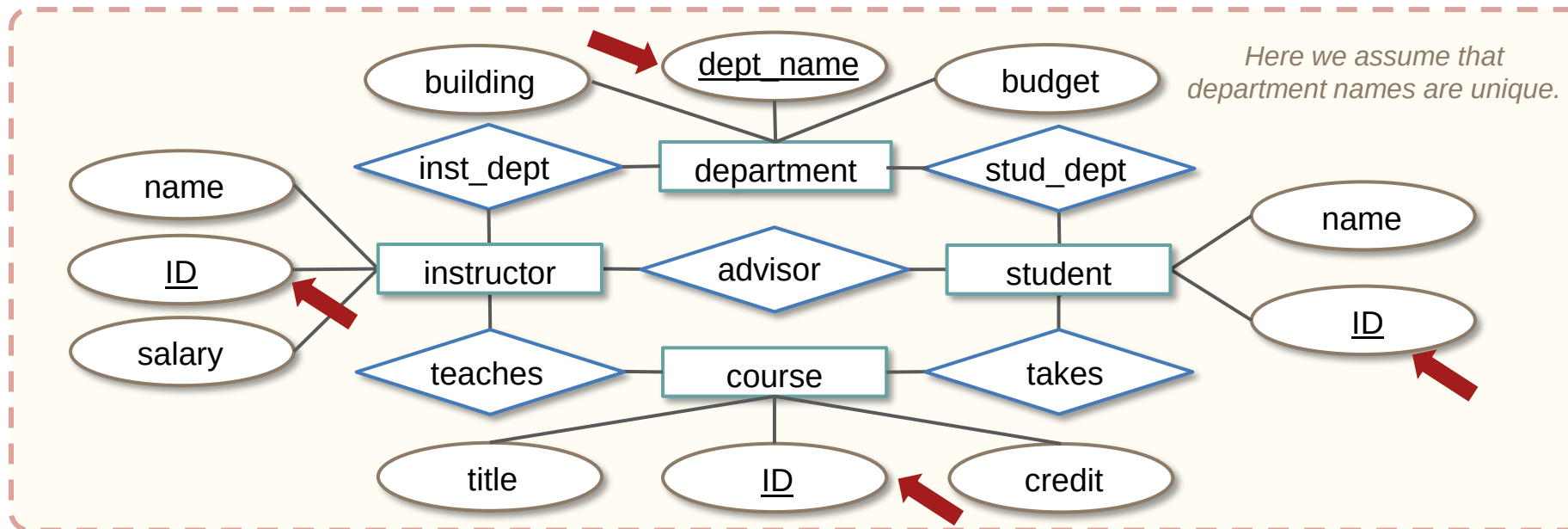
- Age can be derived from a stored birth-date



Attributes



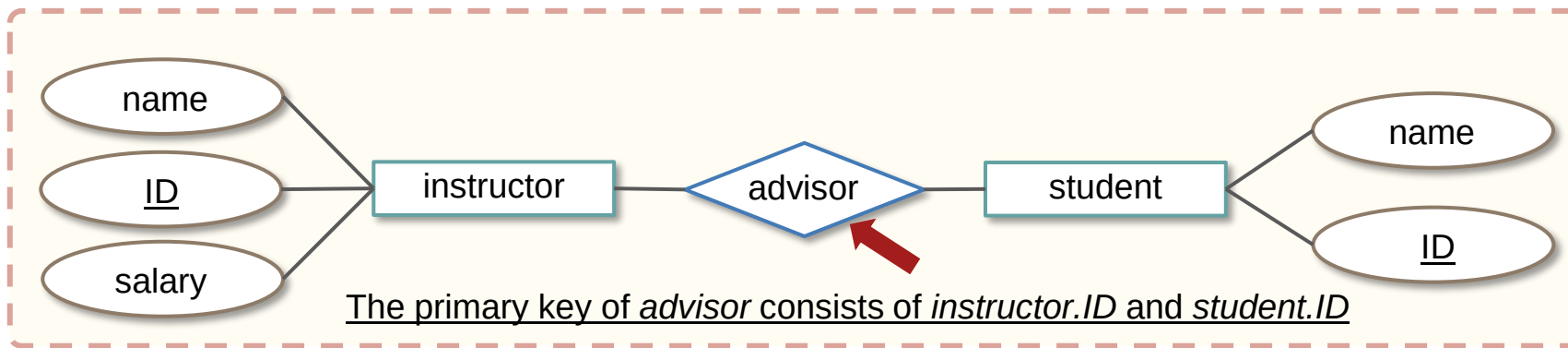
- Primary key for entity sets (same as in the relational model)
 - Concept
 - ◆ An attribute, or a minimal set of attributes, that uniquely identifies each entity within an entity set.
 - ER diagram notation
 - ◆ Primary key attributes are marked by underlining.
 - ◆ Example:



Attributes



- Primary key for relationship sets
 - Concept
 - ◆ To distinguish among the various relationships of a relationship set, we use the individual primary keys of the entities in the relationship set.
 - ◆ Let R be a relationship set involving entity sets $E_1, E_2, \dots, E_n$
 - ◆ The primary key for R consists of **the union of the primary keys** of entity sets $E_1, E_2, \dots, E_n$
 - ◆ If the relationship set R has attributes $a_1, a_2, \dots, a_m$ associated with it, then the primary key of R also includes the attributes $a_1, a_2, \dots, a_m$
 - ER diagram notation
 - ◆ *The primary key of a relationship set is usually not shown explicitly, since it is implied by the primary keys of the participating entities.*



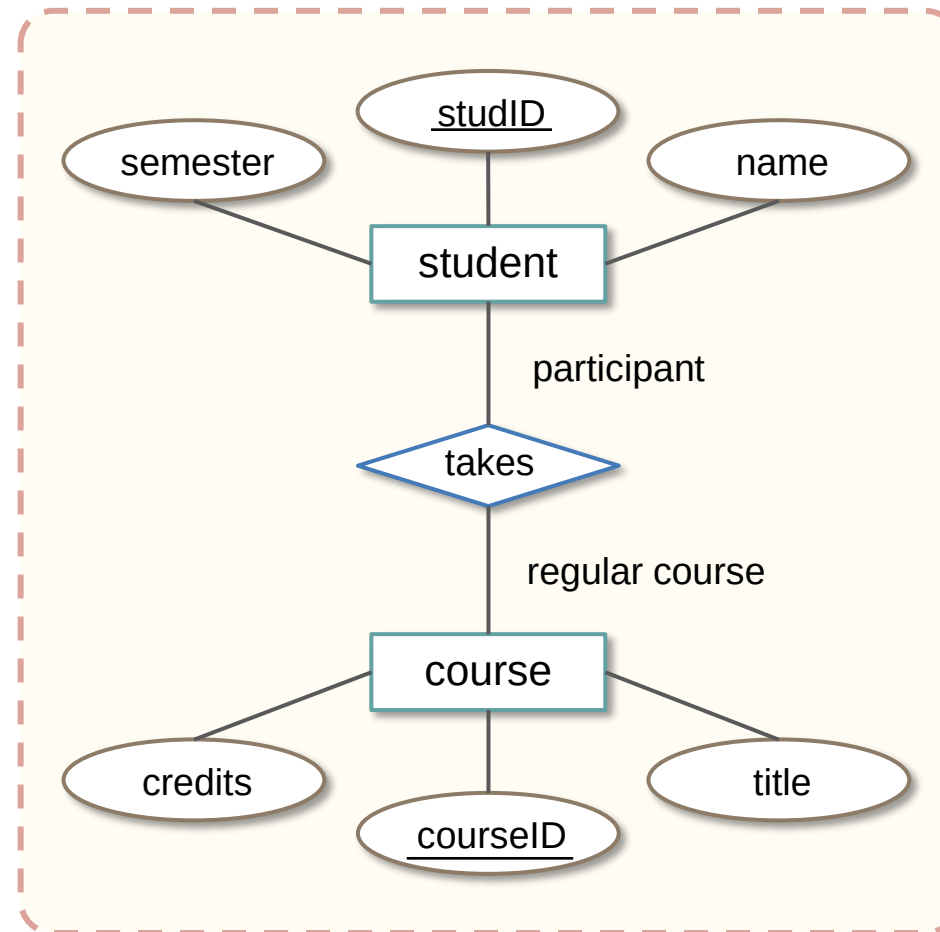
To be discussed later...

The choice of the primary key for a relationship set depends on the **cardinality ratio of the relationship set**.

Example: From Requirements to ER Diagram



- Requirement statement:
 - Students take courses. Each student has a student ID, name, and semester. Each course has a course ID, title, and number of credits. We want to record which students participate in which courses.
- *Step 1: Identify entities*
 - student, course → become entity types
- *Step 2: Identify attributes*
 - student → {studID, name, semester}
 - course → {courseID, title, credits}
- *Step 3: Identify primary keys*
 - studID uniquely identifies a student
 - courseID uniquely identifies a course
- *Step 4: Identify relationships*
 - “take” is an action → relationship type between Student and Course
- *Step 5: Identify roles (if needed)*
 - Student plays role “participant”
 - Course plays role “regular course”



Exercise 1

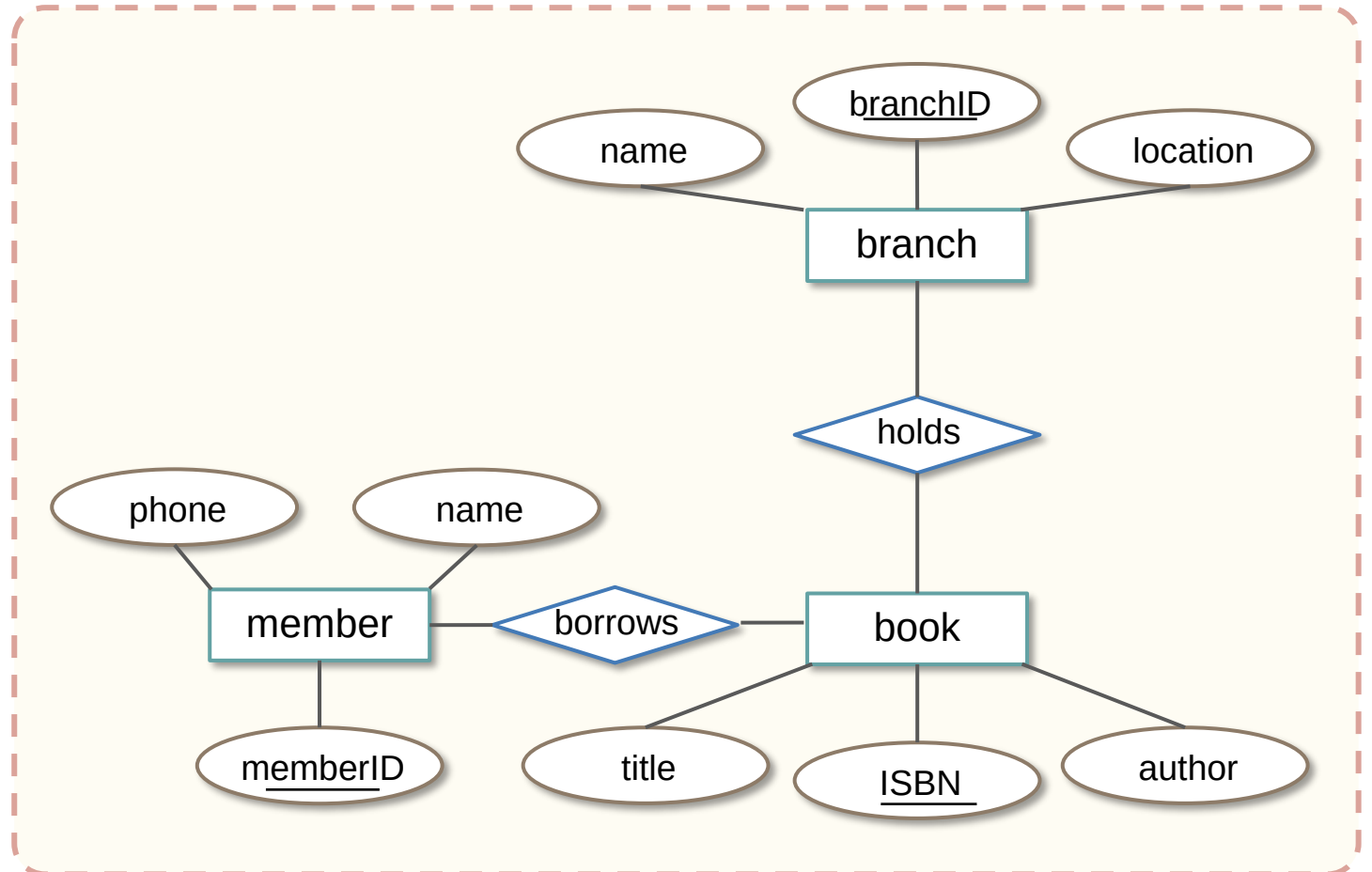


- *Question. Based on the following requirements, construct an ER diagram for a library database. (You may ignore roles in relationships.)*
- *Requirements:*
 - *A library has several branches. Each branch has a name, location, and a unique branch ID.*
 - *Each branch holds many books*
 - *A book has a title, author, and ISBN (unique for each book).*
 - *Members of the library can borrow books.*
 - *A member has a unique member ID, a name, and a phone number.*

Exercise 1 -- Solution



- Requirement statement:
 - A library has several branches. Each branch has a name, location, and a unique branch ID. Each branch holds many books. A book has a title, author, and ISBN (unique for each book). Members of the library can borrow books. A member has a unique member ID, a name, and a phone number.
- *Step 1: Identify entities*
 - branch, book, member
- *Step 2: Identify attributes*
 - branch → {branchID, name, location}
 - book → {ISBN, title, author}
 - member → {memberID, name, phone}
- *Step 3: Identify primary keys*
 - branch: branchID
 - book: ISBN
 - member: memberID
- *Step 4: Identify relationships*
 - “holds”: between branch and book
 - “borrows”: between member and book



Agenda



- Review Steps of Database Design
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Cardinality ratio

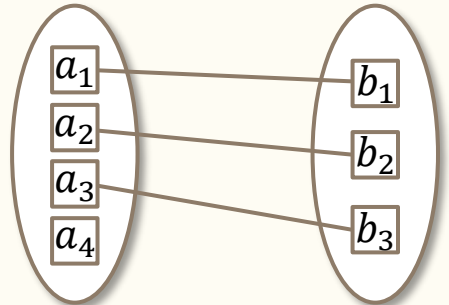


- Concept

- Number of entities involved in a relationship instance
- For a binary relationship set R between entity sets A and B , the mapping cardinalities must be one of the following:

One-to-one(1:1)

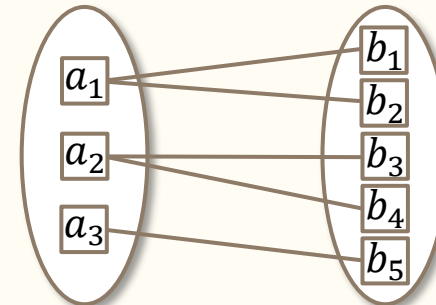
- An instructor advise **at most one** student
- A student have **at most one** advisor.



A (instructor) B (student)

One-to-many(1:N)

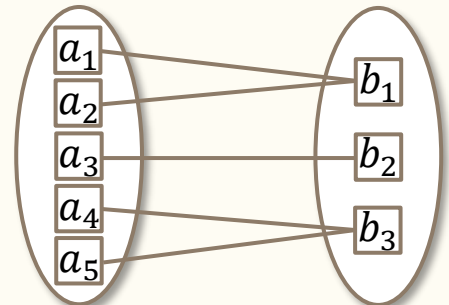
- An instructor may advise **many** students.
- A student have **at most one** advisor.



A (instructor) B (student)

Many-to-one(N:1)

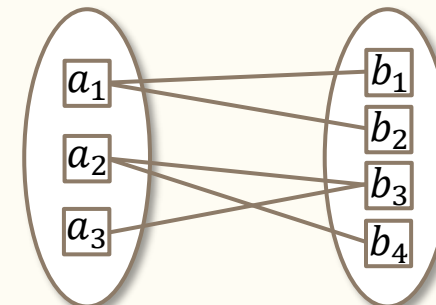
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A (instructor) B (student)

Many-to-many(N:M)

- An instructor may advise **many** students.
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A (instructor) B (student)

Note: Some elements in A and B may not be mapped to any elements in the other set

Cardinality ratio

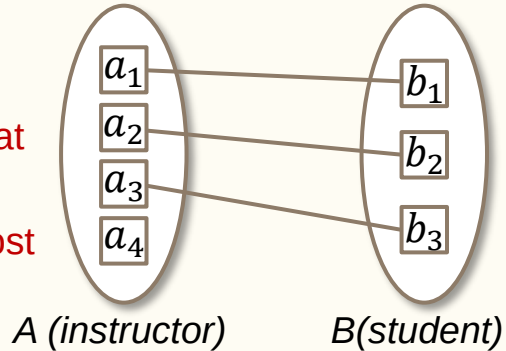


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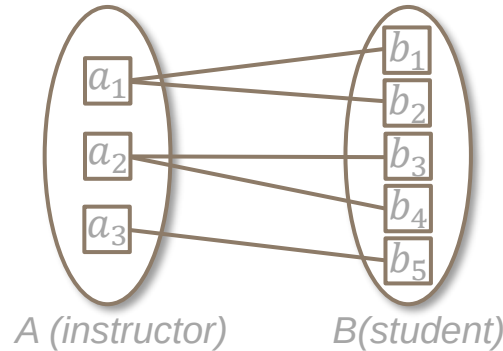
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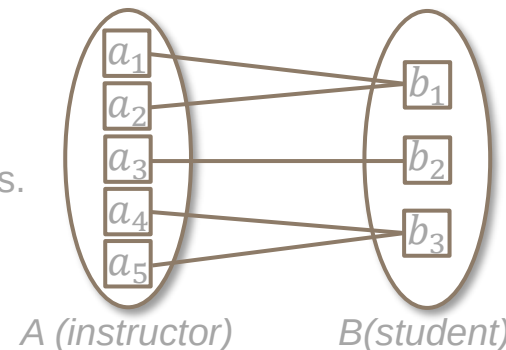
One-to-many

- An instructor may advise many students.
- A student have at most one advisor.



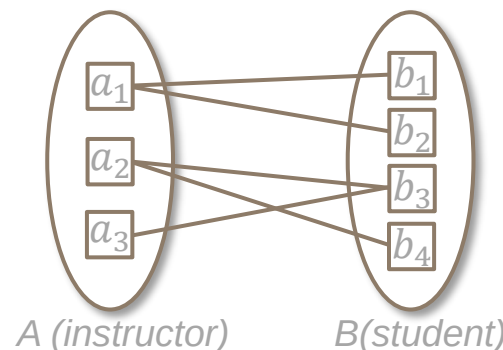
Many-to-one

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Many-to-many

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Cardinality ratio

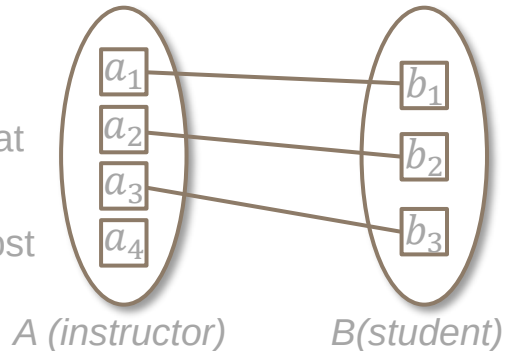


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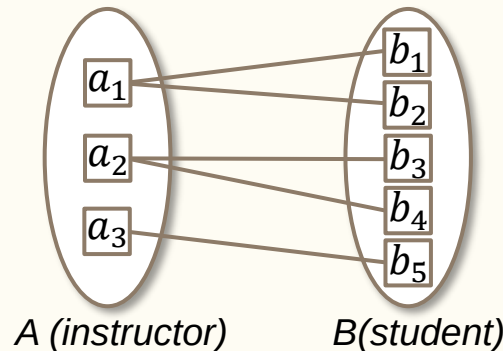
One-to-one

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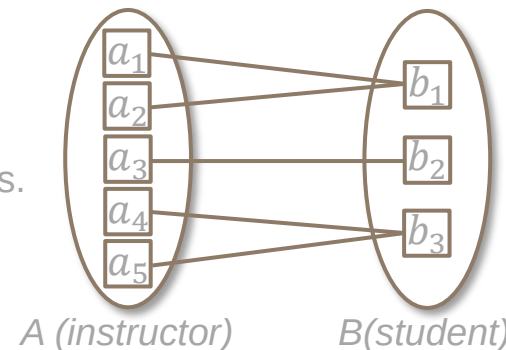
One-to-many(1:N)

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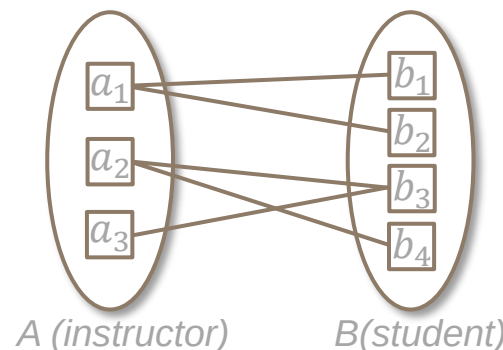
Many-to-one

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Many-to-many

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Cardinality ratio

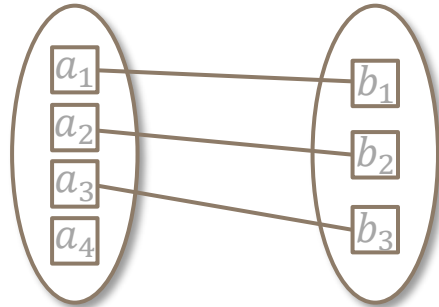


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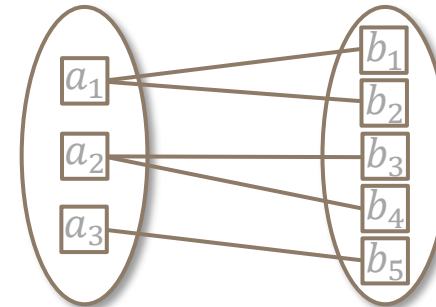
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One-to-many

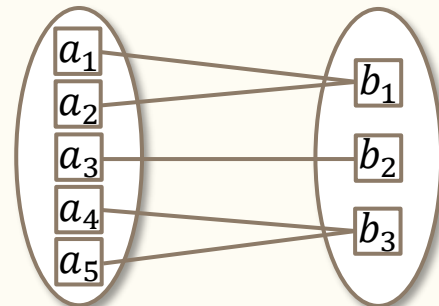
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A (instructor) B (student)

Many-to-one(N:1)

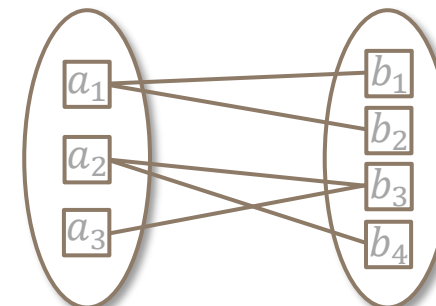
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Many-to-many

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Cardinality ratio

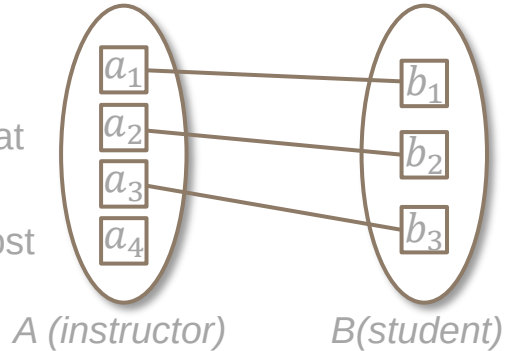


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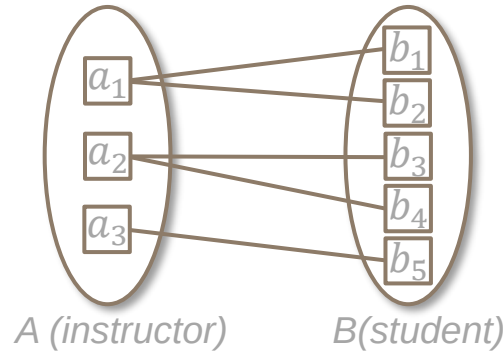
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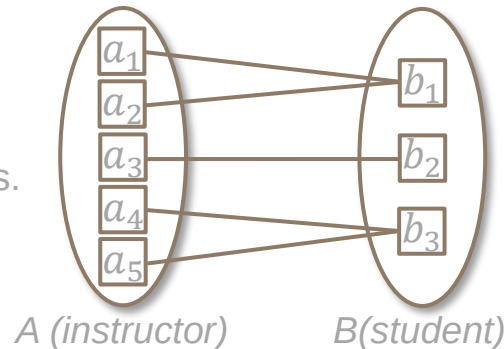
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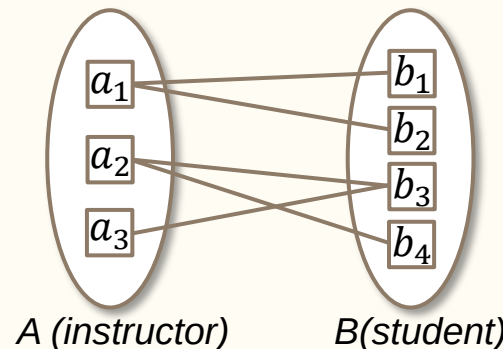
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Many-to-many(N:M)

- An instructor may advise **many** students.
- A student may have **many** advisor.



Note: Some elements in A and B may not be mapped to any elements in the other set

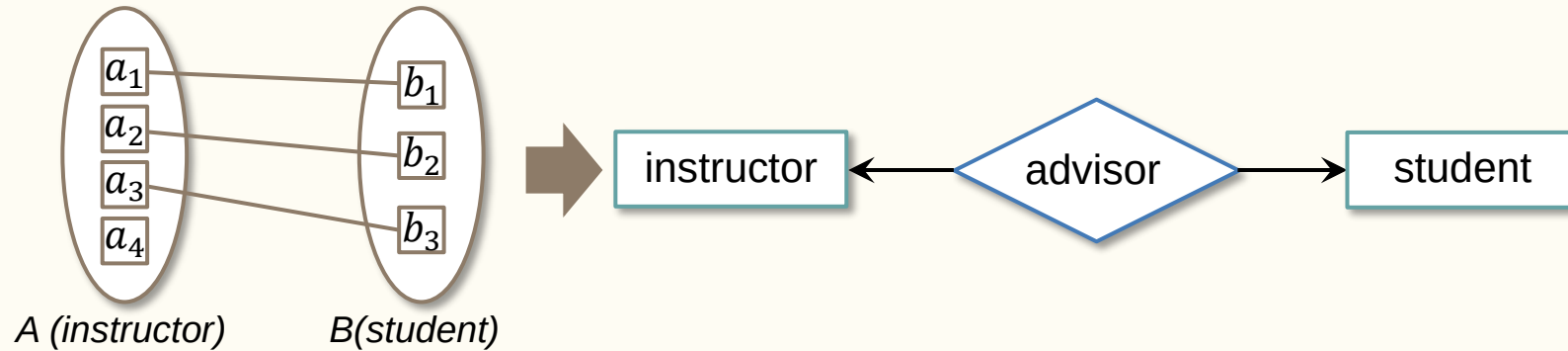
Cardinality ratio



- ER diagram notation for binary relationships (graphical notation)
 - One : a directed line ($\longrightarrow$)
 - Many: an undirected line (---)

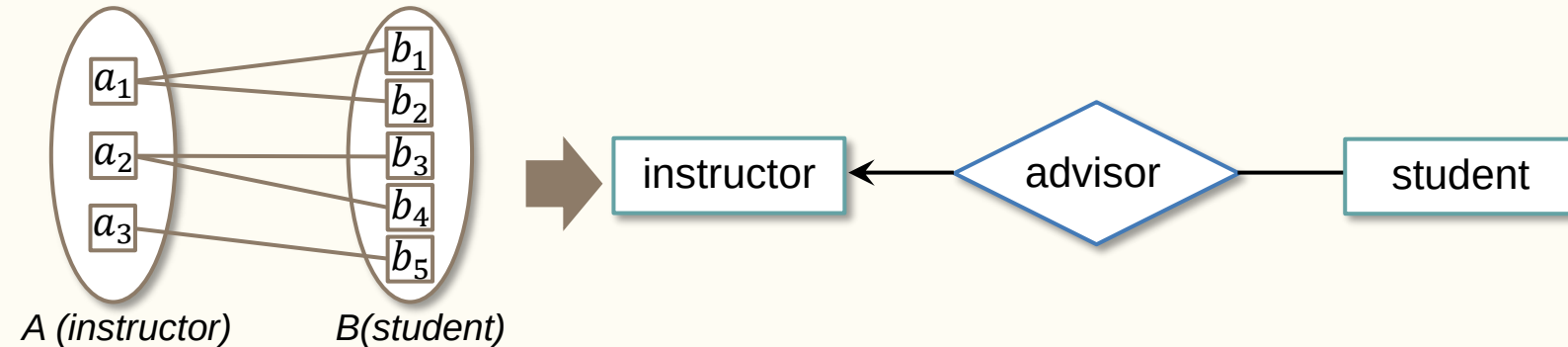
One-to-one(1:1)

- An instructor advise **at most one** student
- A student have **at most one** advisor.



One-to-many(1:N)

- An instructor may advise **many** students.
- A student have **at most one** advisor.



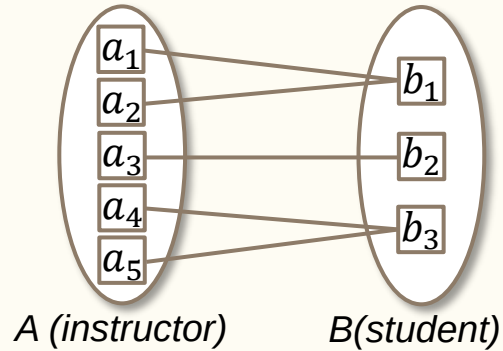
Cardinality ratio



- ER diagram notation for binary relationships (graphical notation)
 - One : a directed line ($\longrightarrow$)
 - Many: an undirected line (---)

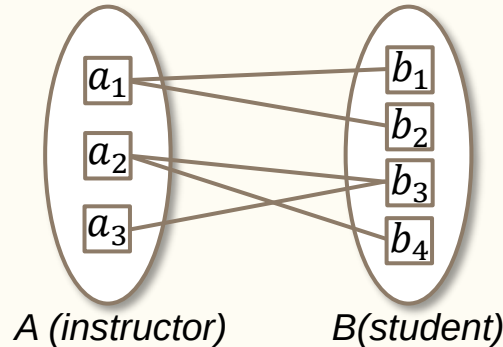
Many-to-one(N:1)

- An instructor advise **at most one** students.
- A student may have **many** advisor.



Many-to-many(N:M)

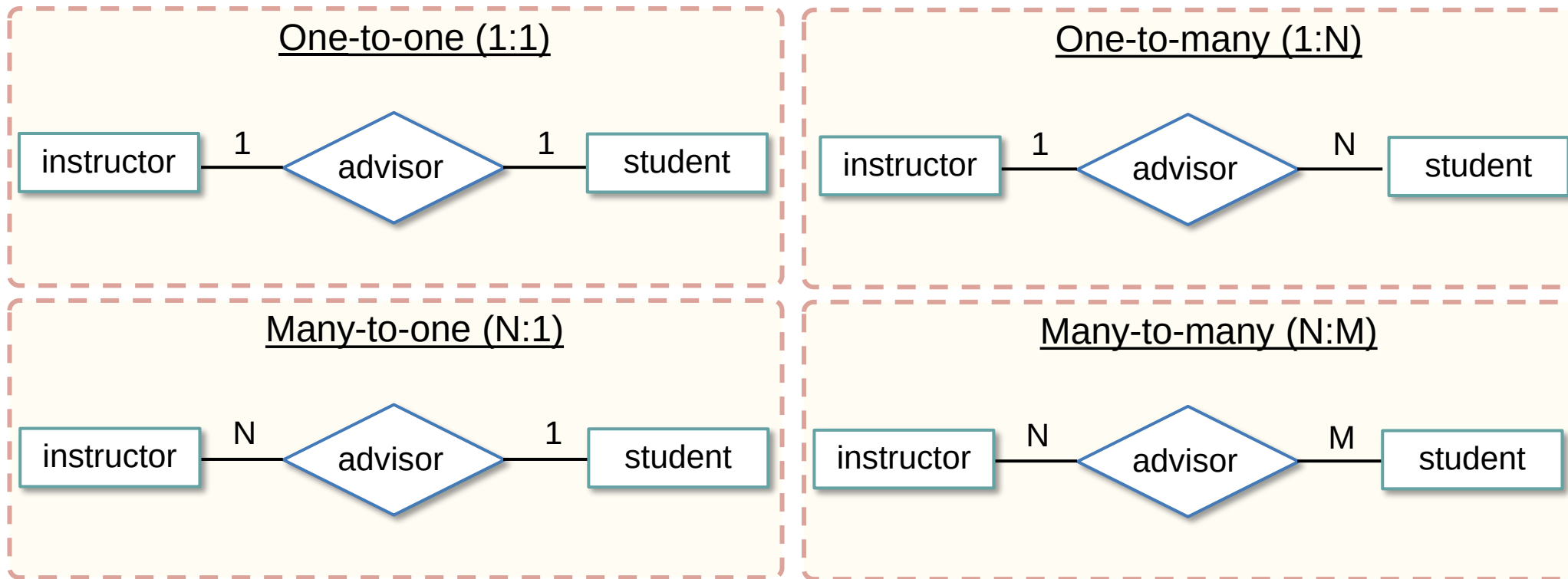
- An instructor may advise **many** students.
- A student may have **many** advisor.



Cardinality ratio



- ER diagram notation for binary relationships (Chen notation)
 - One : mark “1” on the connecting line
 - Many: mark “N” (or M, etc.) on the connecting line

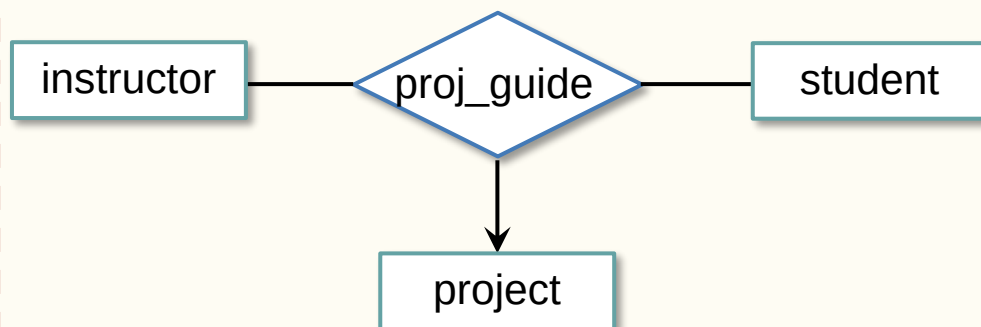


Cardinality ratio

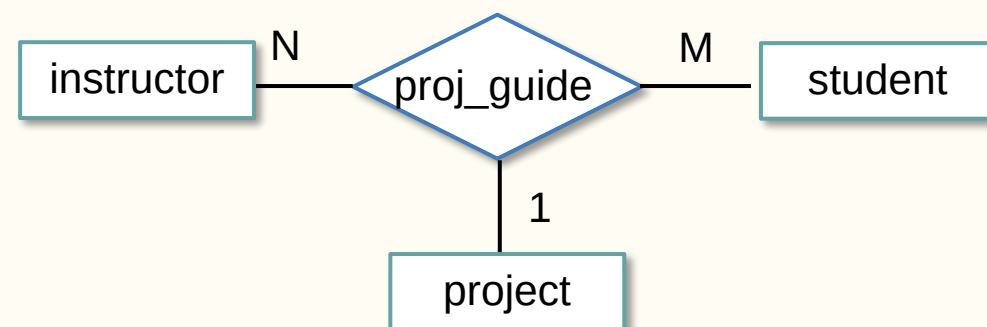


- ER diagram notation for **ternary relationships**
 - Using arrows or annotating lines with 1, N, M, etc. is equivalent.
 - The number or arrow on one edge means:
 - ◆ If the other two entities are fixed, how many instances of this entity can participate in the relationship.
 - Example:
 - ◆ Each project is associated with exactly one instructor–student pair.
 - ◆ Each instructor can guide many students on different projects.
 - ◆ Each student can work on many projects, possibly under different instructors.

Graphical notation



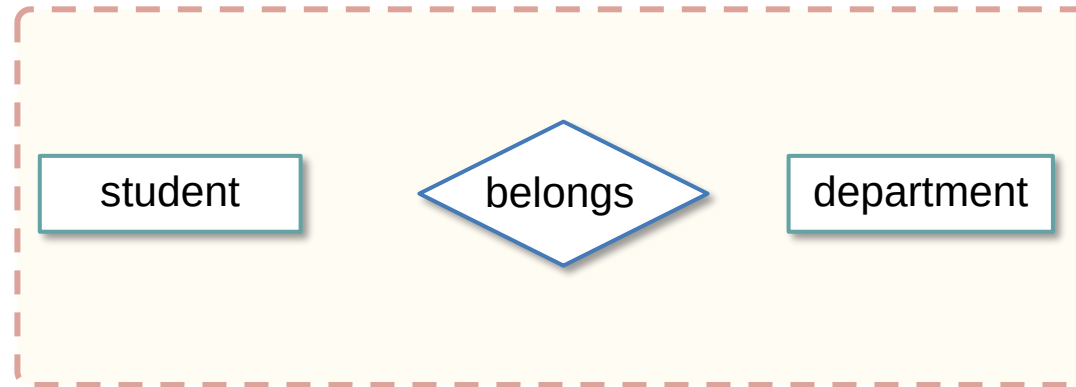
Chen notation



Exercise 2



- Use Graphical Notation and Chen Notation to represent the relationship.
 - *One student belongs to only one department.*
 - *One department can have many students.*

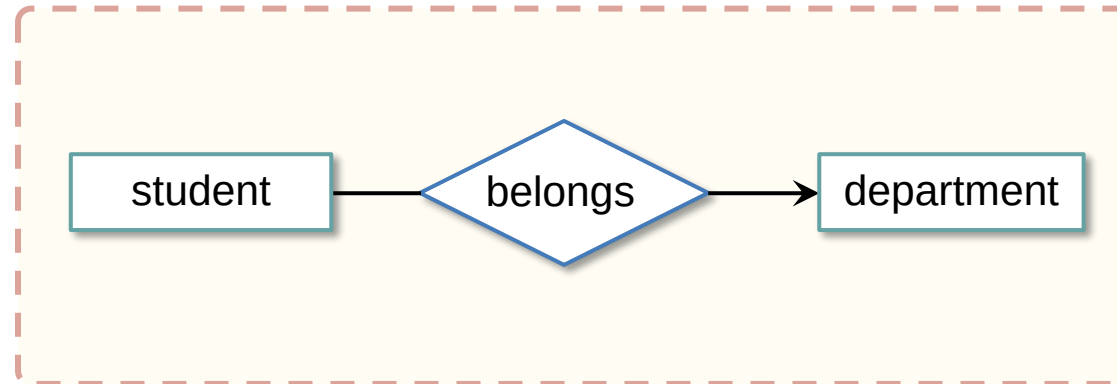


Exercise 2 -- Solution

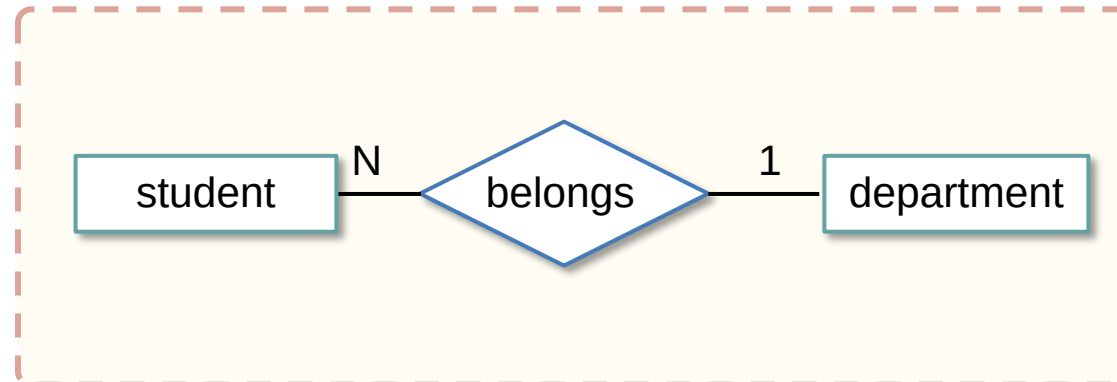


- Use Graphical Notation and Chen Notation to represent the relationship.
 - *One student belongs to only one department.*
 - *One department can have many students.*

Graphical notation



Chen notation



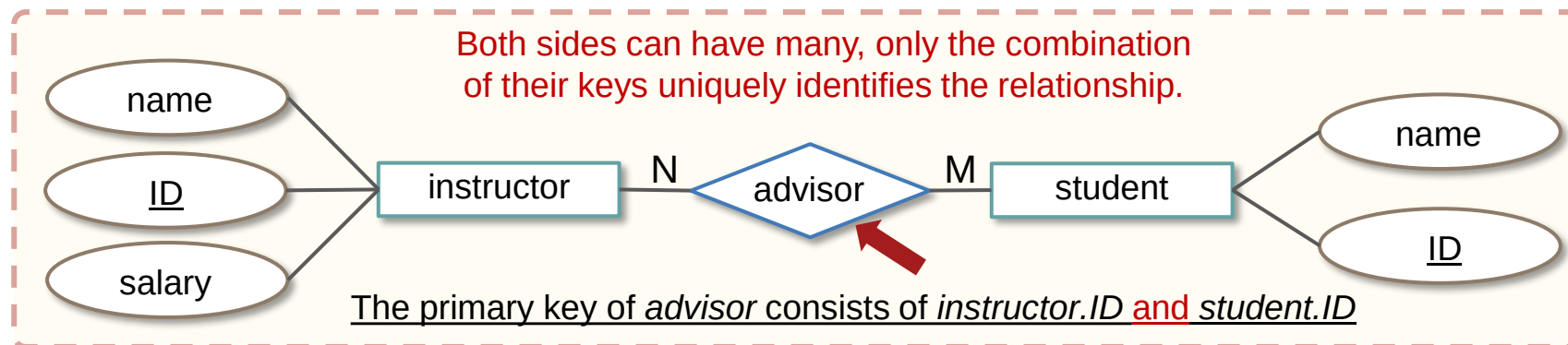
Key Selection from Cardinality Ratios



- Primary key for relationship sets
 - Concept
 - ◆ To distinguish among the various relationships of a relationship set, we use the individual primary keys of the entities in the relationship set.
 - ◆ Let R be a relationship set involving entity sets $E_1, E_2, \dots, E_n$
 - ◆ The primary key for R consists of **the union of the primary keys** of entity sets $E_1, E_2, \dots, E_n$
 - ◆ If the relationship set R has attributes $a_1, a_2, \dots, a_m$ associated with it, then the primary key of R also includes the attributes $a_1, a_2, \dots, a_m$

*The choice of the primary key for a relationship set depends on the **cardinality ratio of the relationship set**.*

- Many-to-Many relationships
 - ◆ The union of the primary keys of entity sets is chosen as the primary key.
 - ◆ Example



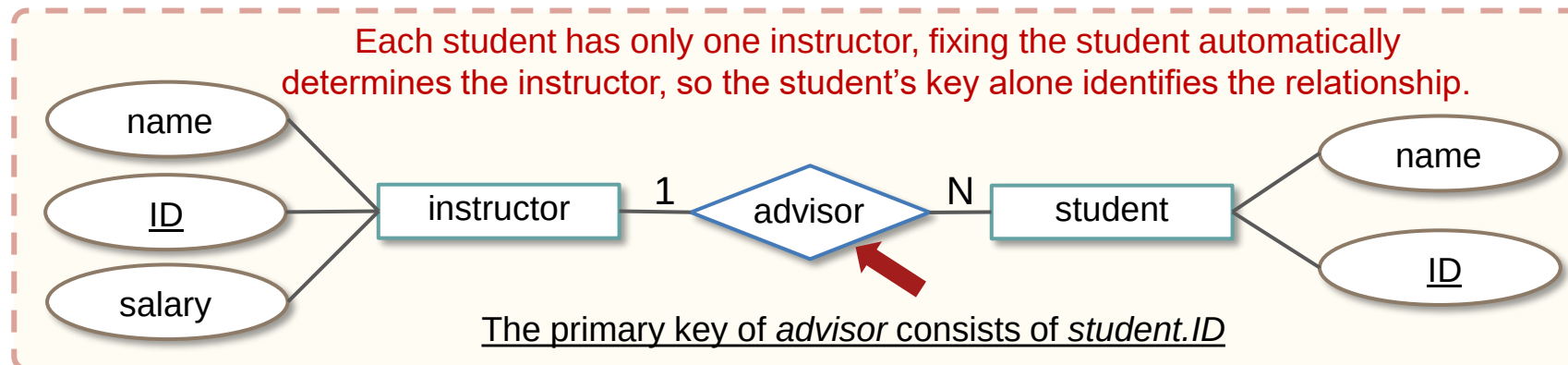
Key Selection from Cardinality Ratios



- Primary key for relationship sets
 - Concept
 - ◆ To distinguish among the various relationships of a relationship set, we use the individual primary keys of the entities in the relationship set.
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*The choice of the primary key for a relationship set depends on the **cardinality ratio of the relationship set**.*

- One-to-Many / Many-to-one relationships
 - ◆ The primary key of the “Many” side is used as the primary key.
 - ◆ Example (One-to-Many)



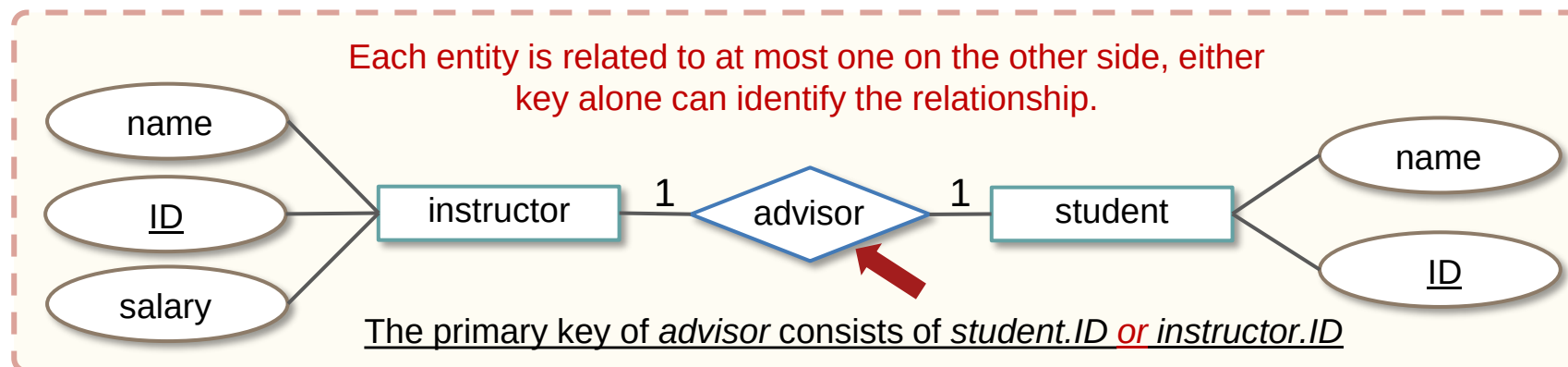
Key Selection from Cardinality Ratios



- Primary key for relationship sets
 - Concept
 - ◆ To distinguish among the various relationships of a relationship set, we use the individual primary keys of the entities in the relationship set.
 - ◆ Let R be a relationship set involving entity sets $E_1, E_2, \dots, E_n$
 - ◆ The primary key for R consists of **the union of the primary keys** of entity sets $E_1, E_2, \dots, E_n$
 - ◆ If the relationship set R has attributes $a_1, a_2, \dots, a_m$ associated with it, then the primary key of R also includes the attributes $a_1, a_2, \dots, a_m$

*The choice of the primary key for a relationship set depends on the **cardinality ratio of the relationship set**.*

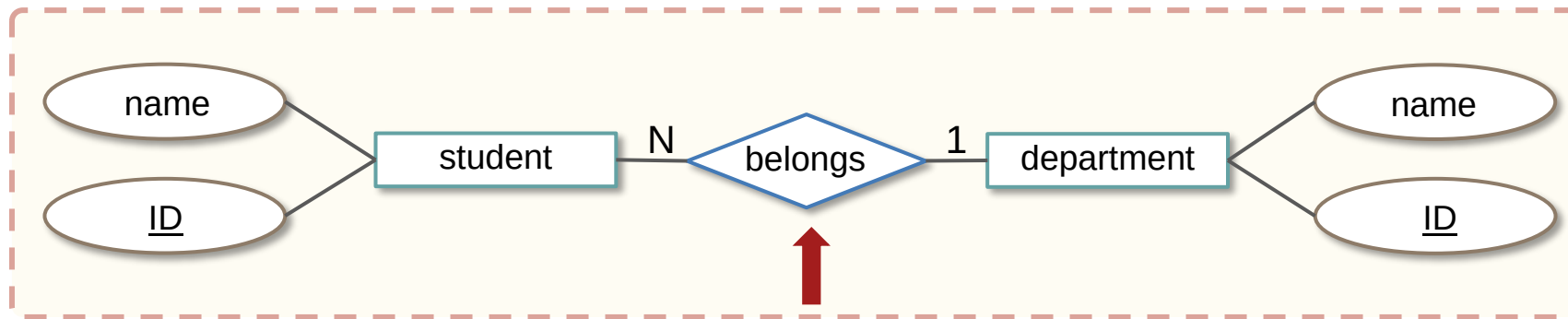
- One-to-one relationships
 - ◆ The primary key of either one of the participating entity sets can be chosen as the primary key.
 - ◆ Example



Exercise 3



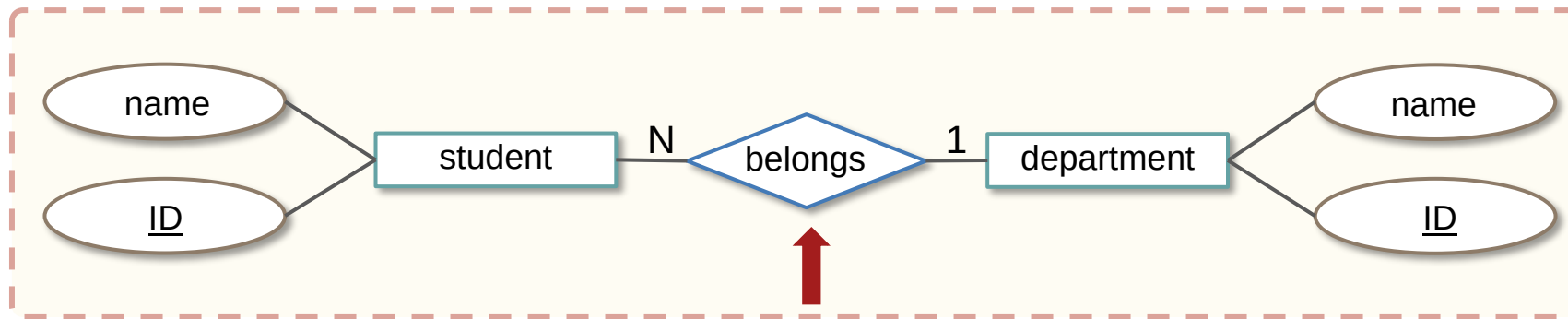
- Which attribute should be the primary key of belongs?
 - *student.ID*
 - *department.ID*
 - *student.name*
 - *department.name*



Exercise 3 --Solution



- Which attribute should be the primary key of belongs?
 - ***student.ID***
 - *department.ID*
 - *student.name*
 - *department.name*

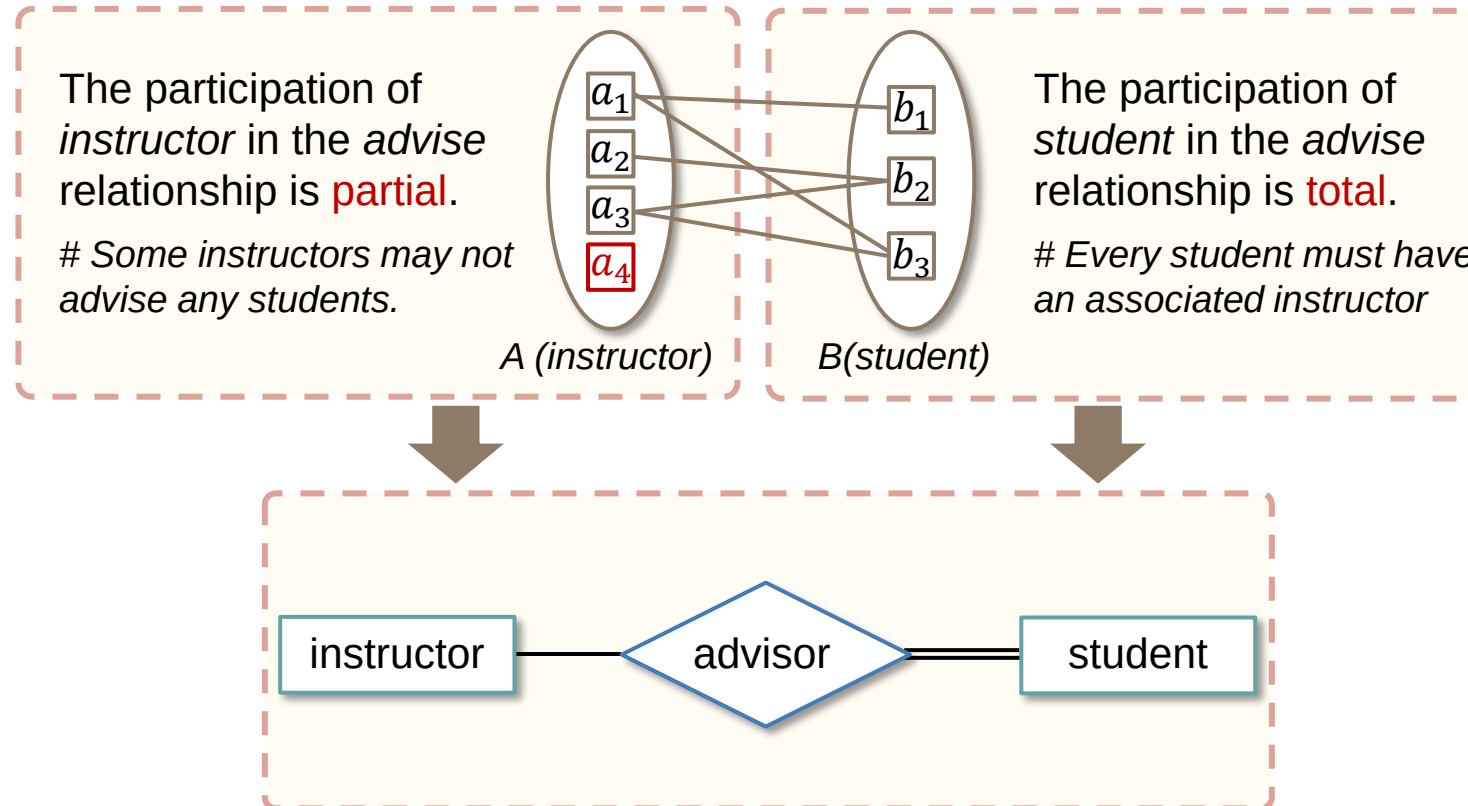


A department can have many students, once the student ID is fixed, the department is determined, so the student's key (student.ID) uniquely identifies the relationship.

Participation constraint



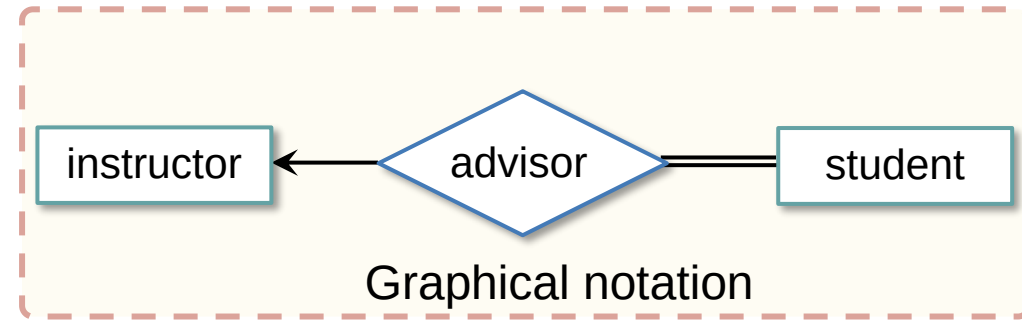
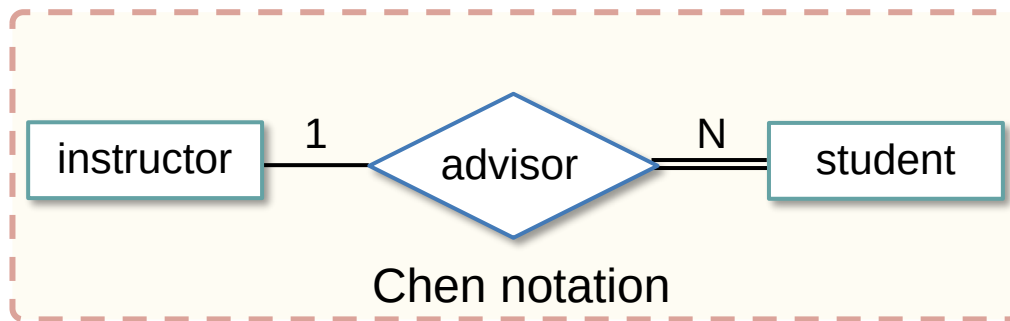
- **Total participation** (indicated by **double line**): every entity in the entity set participates in at least one relationship in the relationship set
- **Partial participation**: some entities may not participate in any relationship in the relationship set



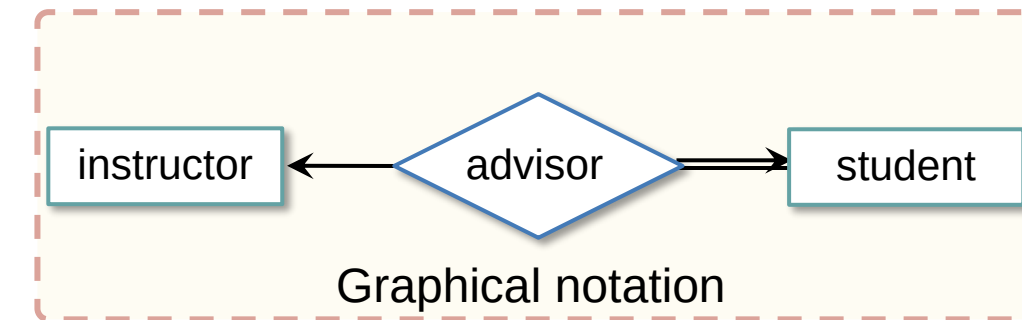
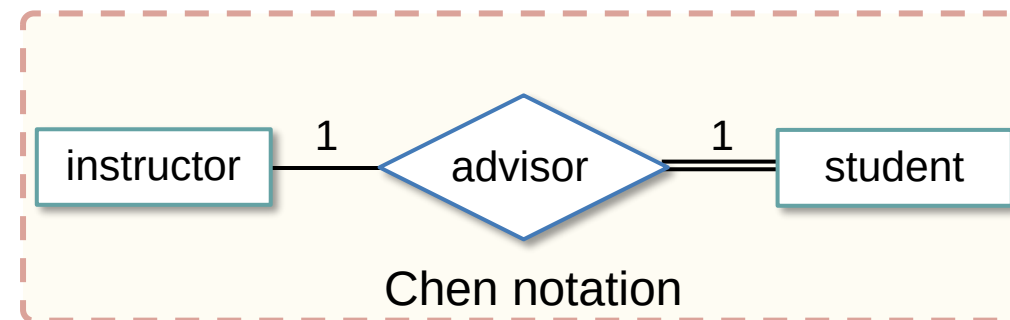
Participation constraint



- Example (with cardinality ratios and participation constraints)
 - Every student must have exactly one instructor. (Total participation on *student*)
 - An instructor may advise many students. (Partial participation on *instructor*)



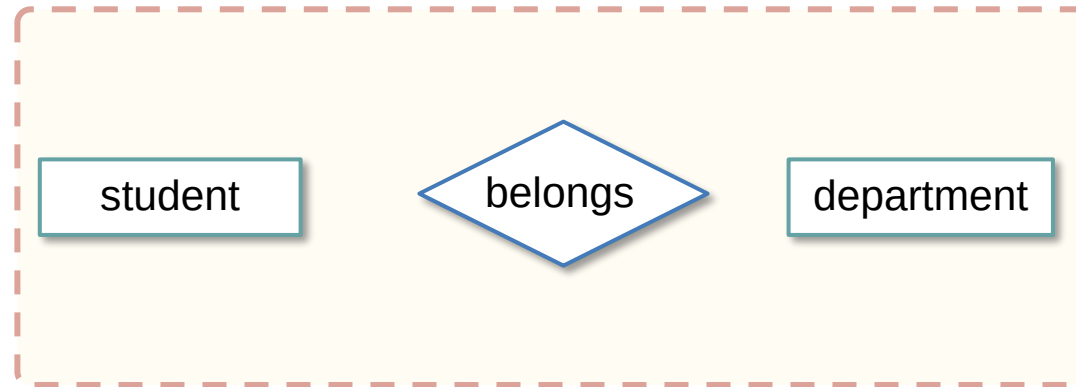
- Every student must have exactly one instructor. (Total participation on *student*)
- An instructor may advise at most one student, but not necessarily. (Partial participation on *instructor*)



Exercise 4



- Use Graphical Notation and Chen Notation to represent the relationship.
 - *Each student must belong to one department*
 - *Each department can have many students or have no students*

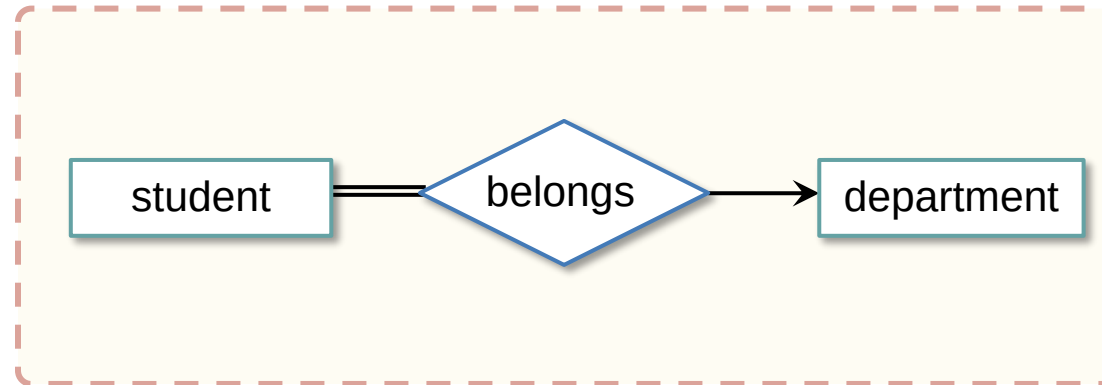


Exercise 4 -- Solution

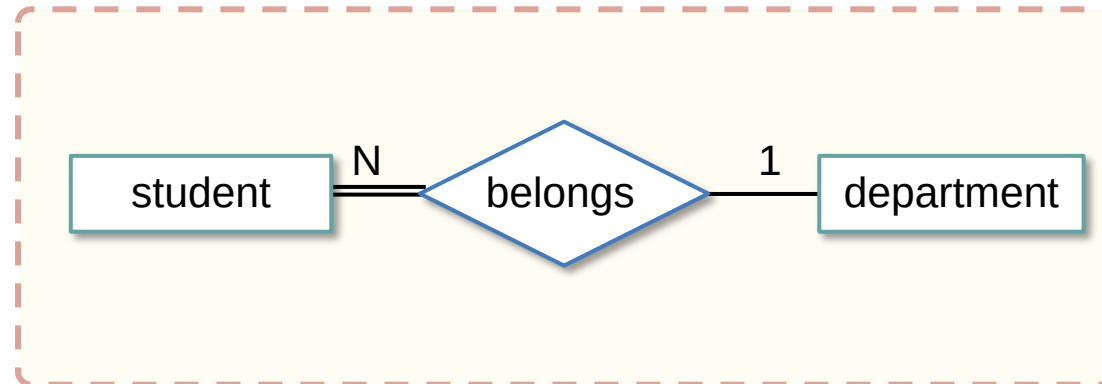


- Use Graphical Notation and Chen Notation to represent the relationship.
 - *Each student must belong to one department*
 - *Each department can have many students or have no students*

Graphical notation



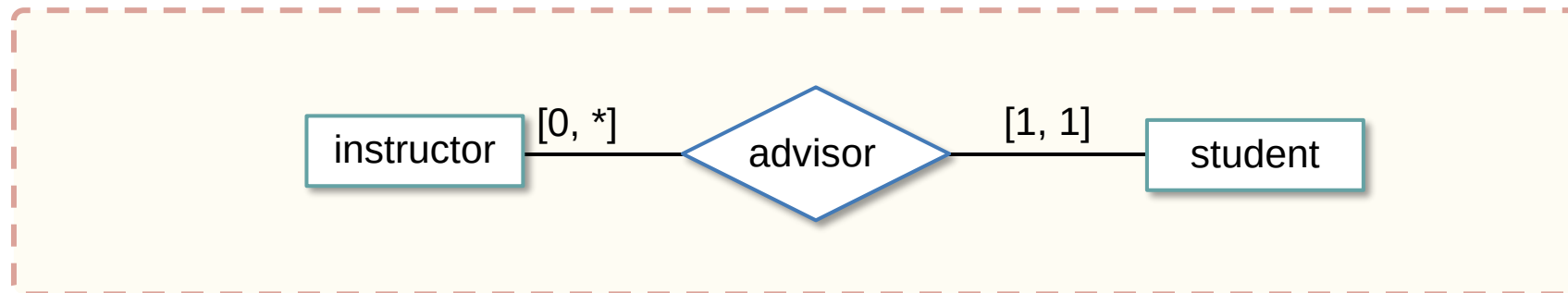
Chen notation



Cardinality Constraints ([min,max] notation)



- A line may have an associated minimum and maximum cardinality, shown in the form [min, max].
 - Minimum: an instance must participate in at least *min* relationships.
 - ◆ If min = 0 → Partial participation (not all instances must participate)
 - ◆ If min ≥ 1 → Total participation (all instances must participate)
 - Maximum: an instance can participate in at most *max* relationships.
 - ◆ * indicates no limit
- Example

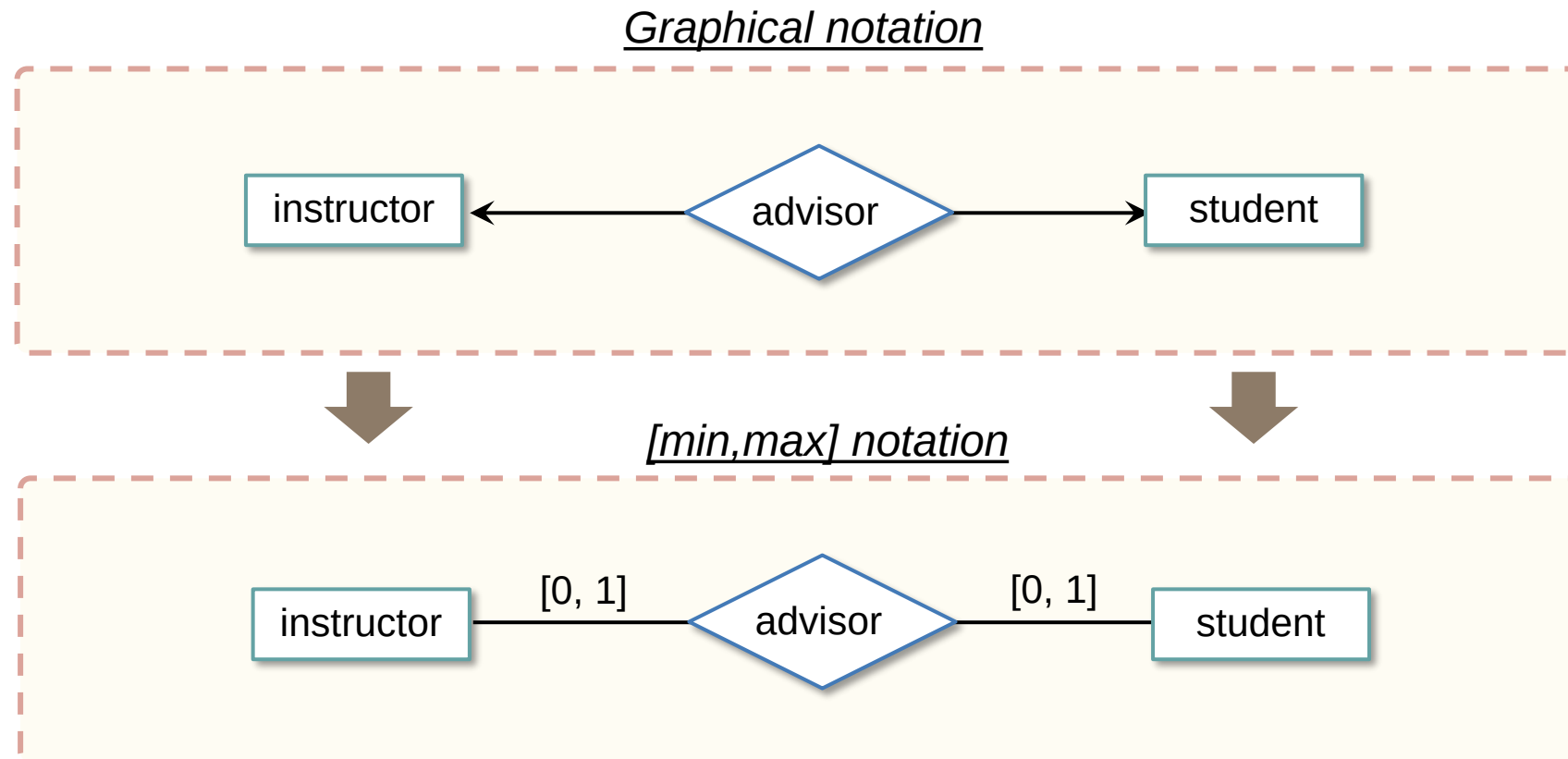


- An instructor can advise 0 or more students.
- A student must have 1 instructor; cannot have multiple instructors.

Cardinality Constraints ([min,max] notation)



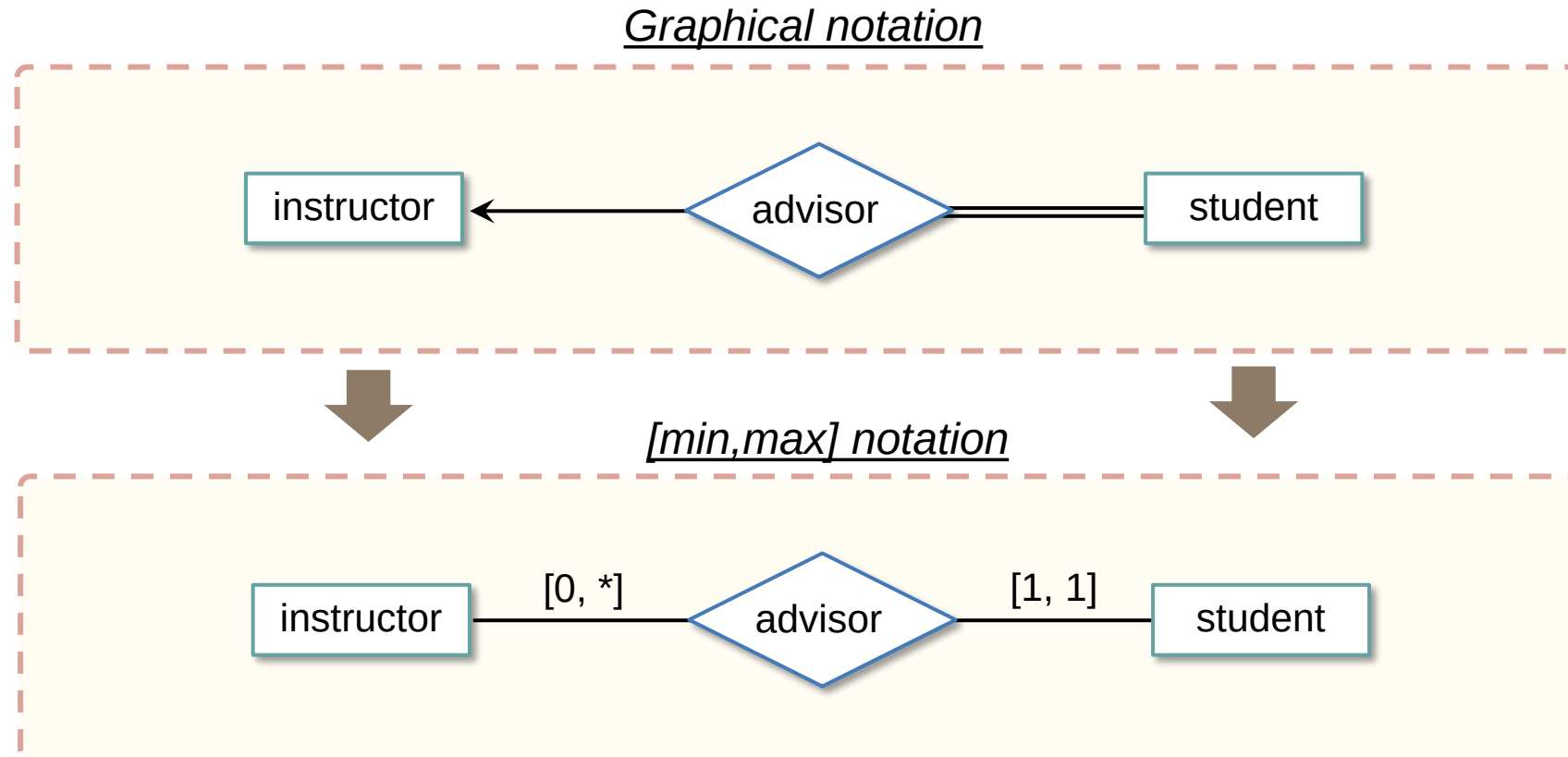
- Example
 - An instructor can supervise at most 1 (0 or 1) student. (Partial)
 - A student can have at most 1 (0 or 1) instructor. (Partial)



Cardinality Constraints ([min,max] notation)



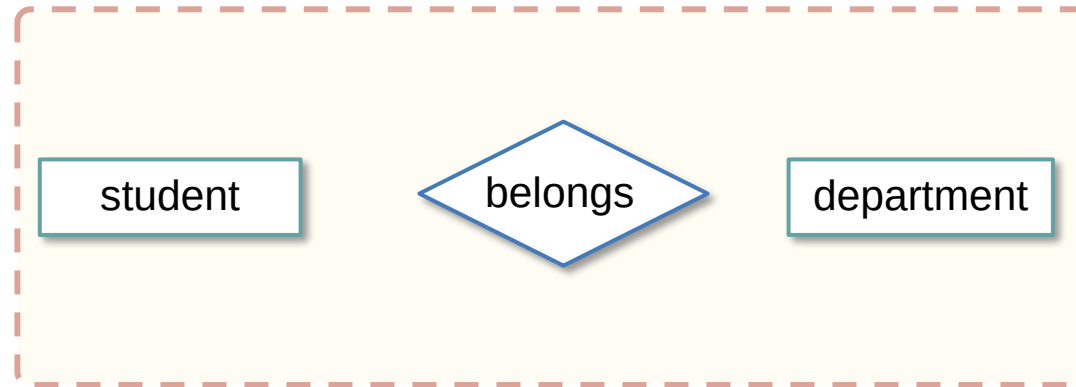
- Example
 - An instructor can supervise 0 or more students. (Partial)
 - A student must have 1 instructor. (Total)



Exercise 5



- Use [min, max] notation to represent the relationship.
 - *Each student must belong to one department*
 - *Each department can have many students or have no students*

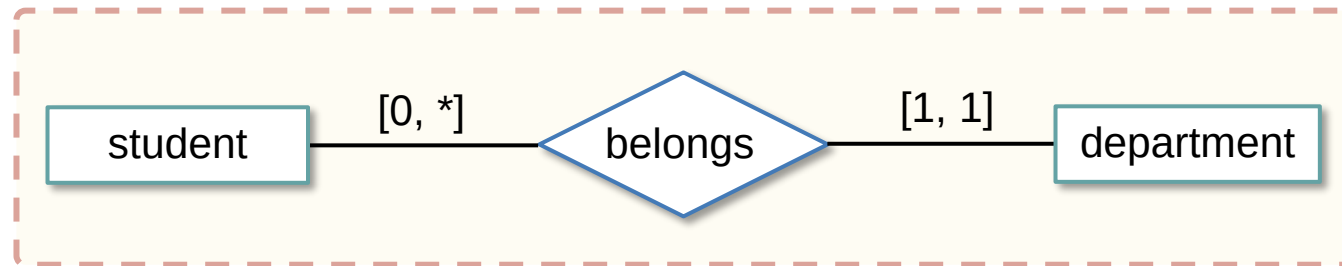


Exercise 5 -- Solution



- Use [min, max] notation to represent the relationship.
 - *Each student must belong to one department*
 - *Each department can have many students or have no students*

[min,max] notation



Agenda

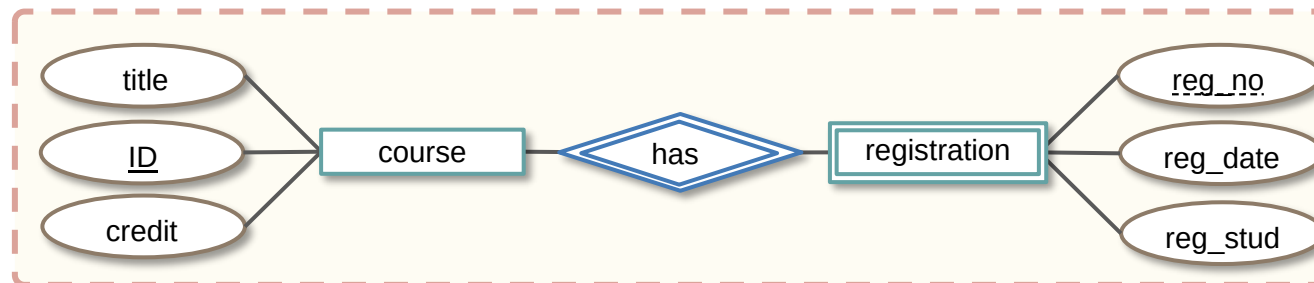


- Review Steps of Database Design
- ER Model
 - Basic Concepts
 - Mapping Cardinality Constraints
- Extended ER Model
 - Weak Entity Types
 - Specialization

Weak Entity Types



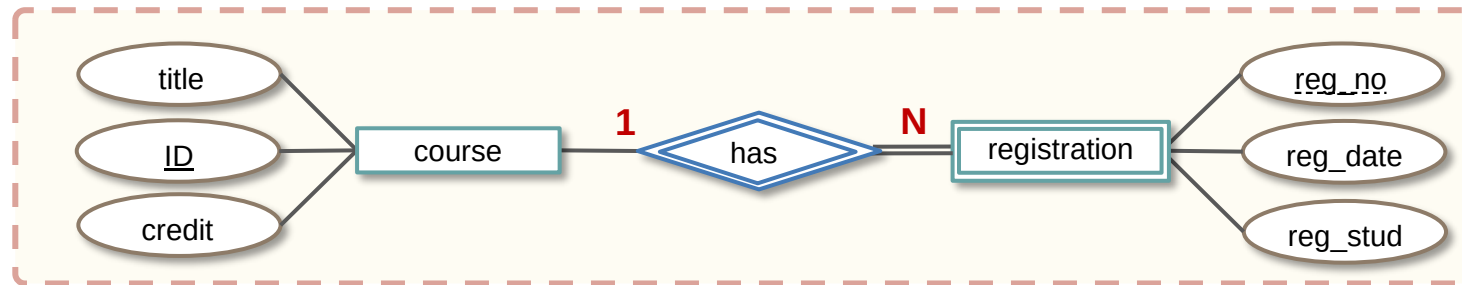
- Concepts
 - A **weak entity type** is one whose existence depends on another (strong) entity, called its **identifying entity**.
 - Weak entities are uniquely identifiable in combination with the corresponding identifying entity's key
 - A **discriminator** is a set of attributes that distinguish weak entities owned by the same identifying entity.
- ER diagram notation
 - Weak entity types → double rectangles
 - Discriminator → underlined with a dashed line
 - Identifying relationships (weak entity ↔ owner) → double diamonds
 - *Regular relationships (even if involving a weak entity) → single diamonds*



Weak Entity Types



- Concepts
 - A **weak entity type** is one whose existence depends on another (strong) entity, called its **identifying entity**.
 - Weak entities are uniquely identifiable in combination with the corresponding identifying entity's key
 - A **discriminator** is a set of attributes that distinguish weak entities owned by the same identifying entity.



- *Total participation of the weak entity type.*
- *Only in combination with 1:N (N:1) relationship types*
- *The identifying entity type is always on the “1”-side*

Agenda



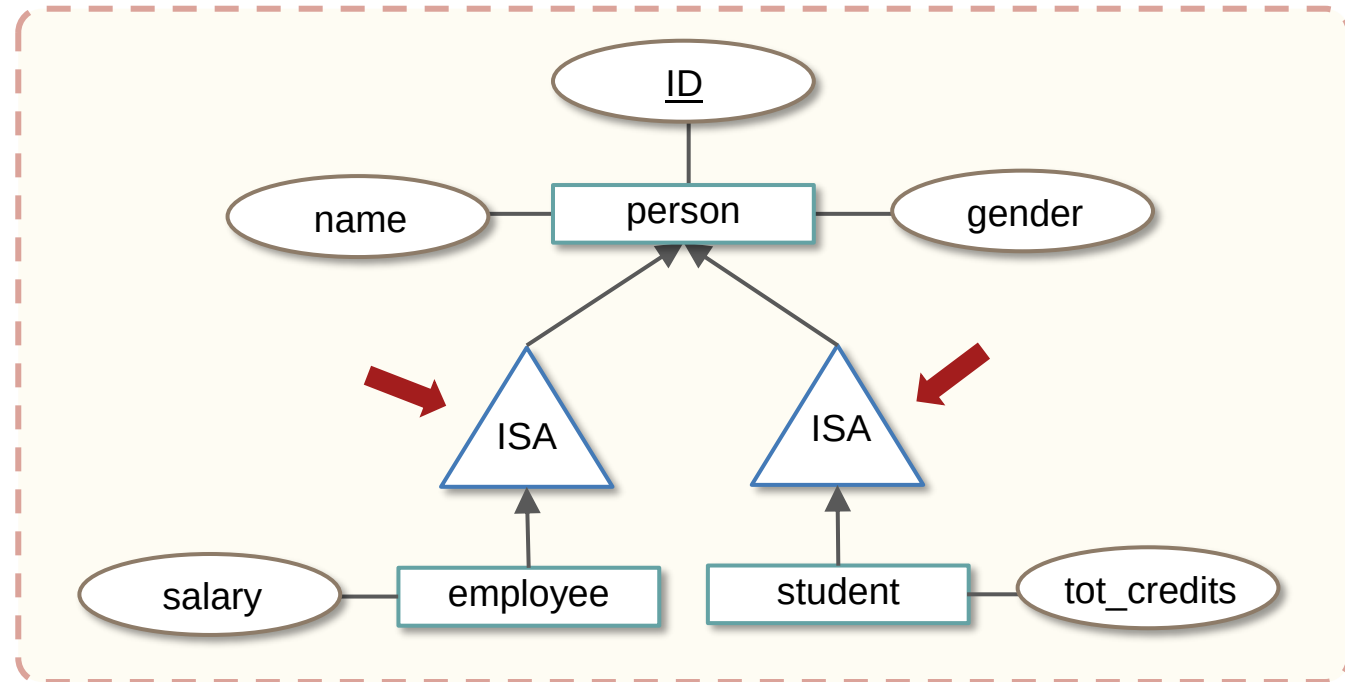
- Review the Relational Model
- ER Model
 - Basic Concepts
 - Mapping Cardinality Constraints
- Extended ER Model
 - Weak Entity Types
 - Specialization

Specialization



- Concept
 - **Specialization** is the process of creating sub-entity sets from a higher-level entity set.
 - It is a *top-down design step*: start with a general entity (e.g., Person), then define more specific sub-entities (e.g., Instructor, Student).

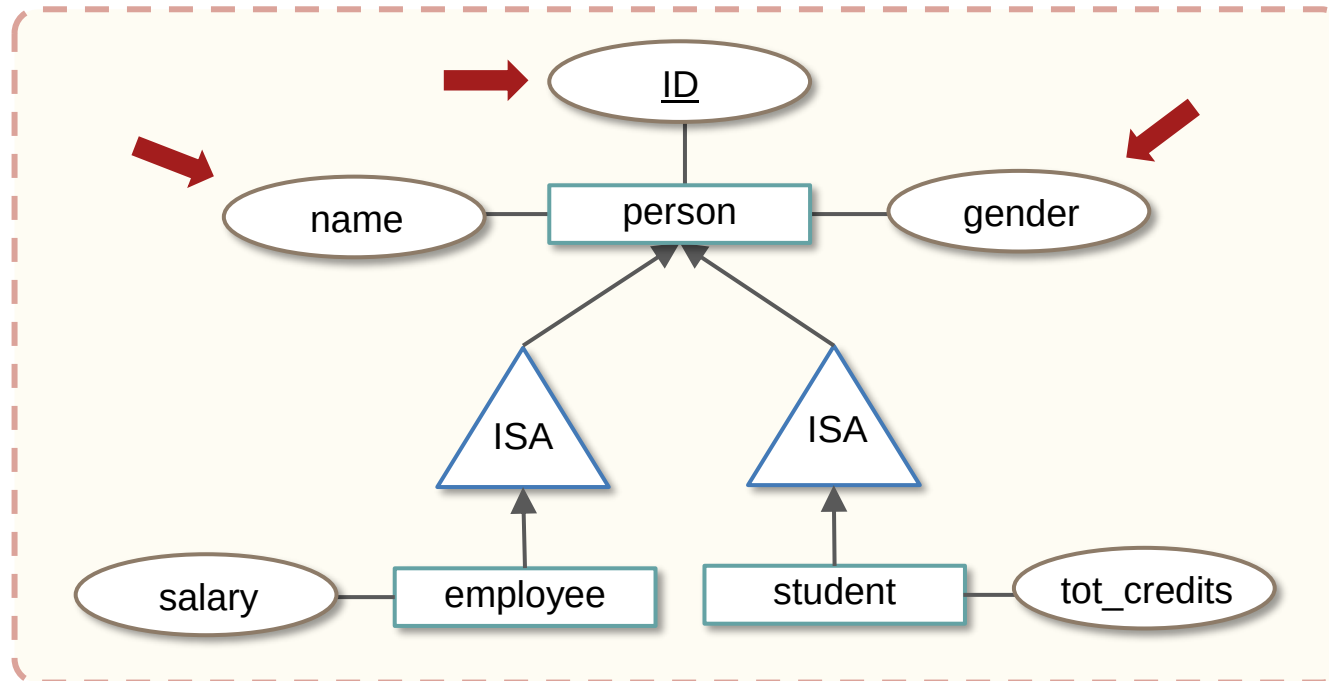
- ER diagram notation
 - A triangle labeled ISA (is a)
 - Example
 - ◆ An employee is a person
 - ◆ A student is a person



Specialization



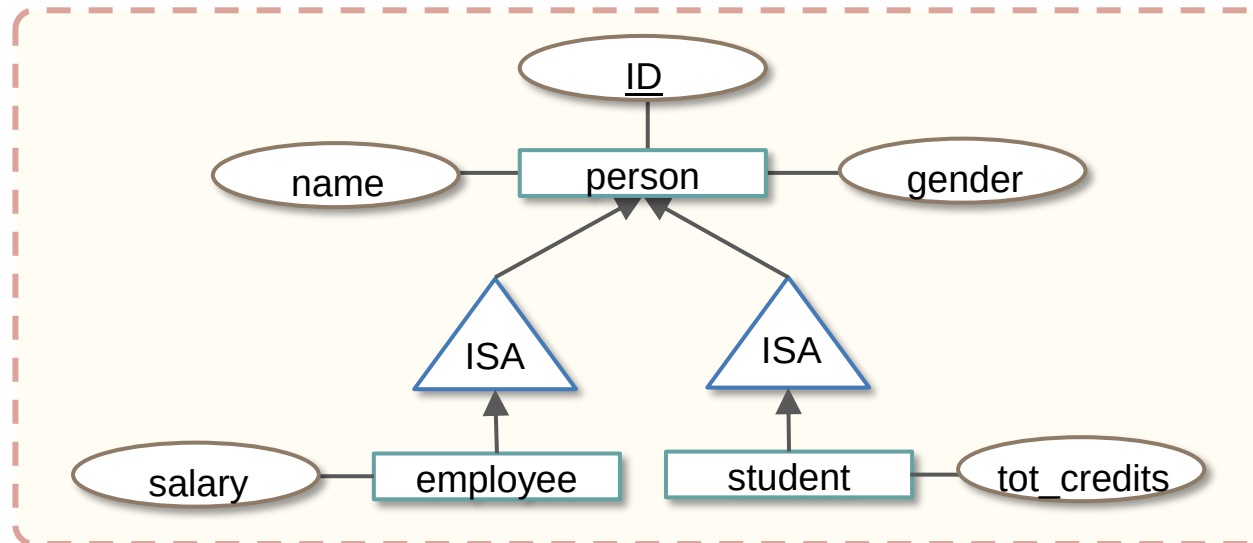
- Attribute inheritance
 - A lower-level entity set inherits all the attributes and relationship participation of the higher-level entity set to which it is linked.
 - Example
 - ◆ *employee* (ID, name, gender, salary)
 - ◆ *student* (ID, name, gender, tot_credits)



Specialization



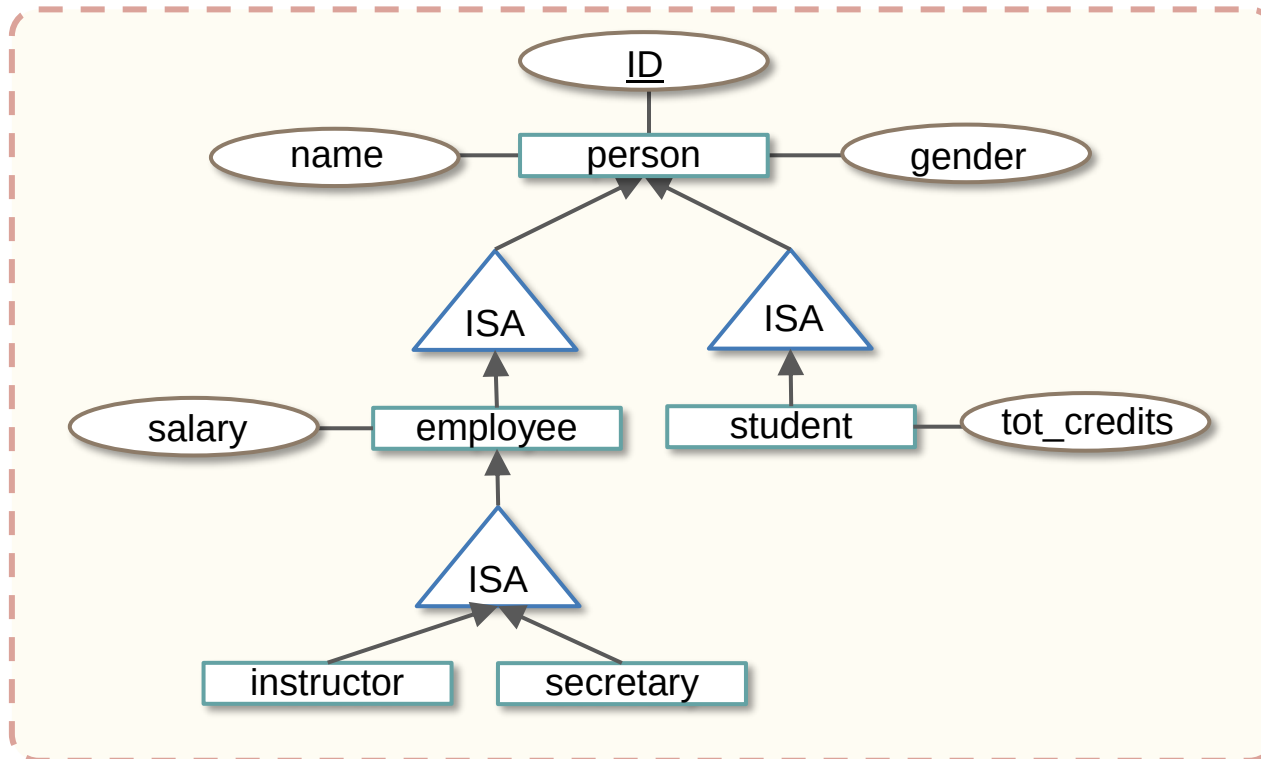
- Completeness constraint
 - Completeness constraint specifies whether or not an entity in the higher-level entity set must belong to at least one of the lower-level entity sets within generalization.
 - **Total specialization** : an entity must belong to one of the lower-level entity sets
 - ◆ Example: Every person in the database is either a student or an employee (no “pure” persons).
 - **Partial specialization**: an entity need not belong to one of the lower-level entity sets
 - ◆ Example: A person in the database could be just a “person” with no student or employee role (e.g., a visitor).



Specialization



- Overlapping specialization
 - An entity may belong to multiple specialized entity sets.
 - Separate ISA symbols are used
- Disjoint specialization
 - An entity may belong to at most one specialized entity set.
 - Arrows to a shared ISA symbol in the diagram



- *Overlapping specialization*
 - *employee and student*
- *Disjoint specialization*
 - *Instructor and secretary*

Summary



- Review Steps of Database Design
- ER Model
 - Basic Concepts
 - Mapping Cardinality Constraints
- Extended ER Model
 - Weak Entity Types
 - Specialization



Any Questions?

Lecture 8

The Entity Relationship Model (Part 1)

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Center for Data-Intensive Systems

Reference



- Abraham et al.. Database System Concepts (seventh edition) – Chapter 6
- Tiantian Liu. Database Systems(SW5) - 5. ER Model (Part 1)
- Christian S. Jensen. Database Systems - The Entity Relationship Model